CAUSAL GRAPH DISCOVERY AND GRAPH-GUIDED MIXTURE-OF-EXPERTS FOR CLIMATE AND REGIONAL WEATHER FORECASTING

A Thesis

Submitted in partial fulfilment for the Degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

by

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Acknowledgements

We express our sincere gratitude to our thesis advisor, Dr. Vinith, [Designation of the Thesis Advisor], Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus, for guidance, critical discussion, and constructive review throughout the course of this work.

We thank Dr. K.P. Soman, Professor and Dean, Department of Artificial Intelligence, for the academic environment and computational facilities of the department, and the faculty members and staff of the department for their reviews and feedback during the project evaluations.

We acknowledge the developers and maintainers of the open-source scientific software on which this work depends, in particular the authors of the Tigramite causal-discovery package, against which our vectorised PCMCI implementation was verified, and the NumPy, SciPy, pandas, scikit-learn, PyTorch, PyTorch Geometric, river and Matplotlib projects. We also acknowledge the agencies and research groups that produce and openly distribute the reanalysis and satellite-derived climate data on which tropical intraseasonal research rests.

Finally, we thank our parents, families and friends for their patience, encouragement and support during our B.Tech programme.

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Abbreviations

ADWIN	–	ADaptive WINdowing drift detector
BH	–	Benjamini–Hochberg false-discovery-rate procedure
BSISO	–	Boreal Summer Intraseasonal Oscillation
DIR	–	Discovering Invariant Rationales
ENSO	–	El Niño–Southern Oscillation
EOF	–	Empirical Orthogonal Function
ERA5	–	Fifth-generation ECMWF reanalysis
FDR	–	False Discovery Rate
GNN	–	Graph Neural Network
h850	–	Geopotential height at 850 hPa
IMDAA	–	Indian Monsoon Data Assimilation and Analysis reanalysis
ISD	–	NOAA Integrated Surface Database
MCI	–	Momentary Conditional Independence
MJO	–	Madden–Julian Oscillation
MoE	–	Mixture of Experts
MSE	–	Mean Squared Error
OLR	–	Outgoing Longwave Radiation
OOD	–	Out-of-Distribution
ParCorr	–	Partial-correlation independence test
PCMCI	–	PC condition selection + Momentary Conditional Independence
PW	–	Precipitable Water
RMSE	–	Root-Mean-Square Error
SST	–	Sea Surface Temperature
STARS	–	Sequential T-test Analysis of Regime Shifts
u200, u850	–	Zonal wind at 200 and 850 hPa

List of Symbols

$R,\ N$	–	Number of regions (50) and of time-series nodes ($N = 6R = 300$)
X_t^i	–	Standardised anomaly of node $i = (r, v)$ (region r, variable v) on day t
Y_{t+L}^r	–	OLR target of region r at lead L days ($L = 7$ or 1)
$\tau,\ \tau_{\max}$	–	Lag in days; maximum lag tested (60)
$\mathcal{S}$	–	Set of BSISO-season target days
$\hat{\mathcal{P}}(X_t^j)$	–	Estimated lagged parents of X_t^j
$\hat{\rho}^{\mathrm{MCI}}$	–	MCI partial correlation (link strength)
q	–	Benjamini–Hochberg adjusted p-value
δ	–	Decoy rate: fraction of surrogate nodes selected as parents
s_e	–	Implied propagation speed of link e (distance / lag)
Π_r	–	Resolution-persistence score of target region r
$\pi_r(r')$	–	Graph prior strength of source region r' for target r
$a_{r'r}$	–	Splitter score of candidate edge $r' \to r$
σ_a	–	Spread of a target's splitter scores (informative if ≥ 0.02)
λ	–	Weight of the variance-across-interventions penalty
$S_{\mathrm{pers}},\ R^2_{\mathrm{clim}}$	–	MSE skill relative to persistence and to climatology
Δ_w	–	Causal-signature distance between consecutive windows

Abstract

Weather and climate regimes shift over time, and forecasting models that choose their predictors by correlation cannot tell whether a dependence is real or merely inherited from autocorrelation and common drivers. Existing graph Mixture-of-Experts methods route by topology or curvature, keep their experts frozen, or rely on hand-constructed shift types; none combines causal structure discovery with drift-triggered expert adaptation for spatiotemporal weather forecasting. This thesis addresses that gap.

Phase 1 forecasts daily outgoing longwave radiation (OLR), a proxy for tropical convection, for 50 tropical regions during the Boreal Summer Intraseasonal Oscillation (BSISO) season. Six variables per region give 300 time-series nodes over 1979–2022, with 1979–2012 for discovery and training and 2013–2022 held out. A vectorised PCMCI implementation, verified against the Tigramite reference, tests 18,000 lagged candidates per target with false-discovery-rate control and finds 536 links. Four independent checks validate the graph: none of 250 phase-randomised decoys is selected, only 3 of 536 links exceed physical propagation-speed limits, a data-driven top-k step fixes the prior at about two source regions per target, and 62% of outside links persist when the regions are redrawn at double resolution. The frozen graph then guides an invariance-trained causal splitter and per-region experts. In the full 50-region candidate pool it turns uninformative source scores into informative ones (0 to 24 regions at a 7-day lead, 0 to 36 at 1 day) and improves 1-day forecast skill in 32 of 50 regions, while 7-day skill is unchanged because the discovered links act within one to two days.

For adaptation, a dual-channel detector fuses an error stream (ADWIN) with a causal-structure stream (STARS). At two of three test regions, the structure channel detected the 1997–98 El Niño shift 2–5 times sooner than the error channel, and a gradual 2013–2022 shift 5–10 times sooner. In the expert lifecycle, 104 of 171 regime changes reactivated an archived expert instead of training from scratch, and blending the three most similar archived experts beat single-expert reactivation on forgetting at every region tested.

Phase 2 carries the validated components into an India-focused hourly forecasting system (IMDAA, ERA5 and station data, up to 72 hours) with a shared graph backbone, causally gated regional experts and benchmarking against GraphCast and WeatherNext, and outlines the path to global scale.

Keywords: causal discovery, PCMCI, BSISO, OLR, mixture of experts, concept drift, expert lifecycle, graph neural networks.

Chapter 1

Introduction

1.1 Background and Motivation

Weather and climate regimes shift over time. An El Niño year does not behave like a neutral year, and the monsoon's intraseasonal rhythm differs from season to season. A forecasting model trained once on historical data therefore goes stale as conditions change. The usual remedy, retraining the entire model from scratch or on a fixed schedule, is wasteful, because a distribution shift is usually confined to particular regions or relationships rather than the whole system. As forecasting systems grow from regional to global scale, the number of regions that can drift independently grows with them, and so does the cost of full retraining.

A second weakness lies in how forecasting systems choose their inputs. Most select source regions by correlation or by geographic proximity. In the tropical atmosphere, correlations are inflated by autocorrelation and by common drivers such as the large convective envelopes of intraseasonal oscillations [1]. A model built on such relationships cannot distinguish "the physics of this region changed" from "a coincidental correlation broke". Causal discovery for time series offers an alternative: it asks whether the past of a source still carries information about a target once the target's other drivers and the source's own memory are accounted for [2–4].

This thesis combines the two ideas. It first discovers and independently validates the causal drivers of each region, and then uses them to guide a Mixture of Experts (MoE) whose experts are adapted only where a measured drift signal says that a region's behaviour has changed.

1.2 The Forecasting Setting: BSISO and OLR

The subseasonal range of two weeks to two months is the least predictable window of weather forecasting and a current research priority [5]. In the tropics its main source of skill is organised, propagating convection: the Madden–Julian Oscillation (MJO) in boreal winter [6] and the Boreal Summer Intraseasonal Oscillation (BSISO) between May and October [7, 8]. The BSISO forms over the equatorial Indian Ocean and moves northward and eastward with a 30–60-day period, driving the active and break spells of the Indian monsoon [9].

The forecast target is outgoing longwave radiation (OLR), the thermal radiation leaving the top of the atmosphere [10]. Deep convective clouds have cold tops and emit little, so low OLR indicates active convection and high OLR indicates suppressed convection [11]. OLR is a convection proxy rather than a rainfall measurement, and it is a demanding target: it has the lowest day-to-day persistence of the six variables used here, so any skill beyond a few days must come from the evolution of the wider system.

1.3 Problem Statement

Can a statistically discovered and independently validated causal graph improve how a Mixture-of-Experts forecaster selects its source regions, and can causal-structure drift detection adapt individual regional experts, rather than the whole model, when the climate regime changes?

1.4 Objectives

Phase 1 (this thesis).

O1. Discover a causal graph, not just a correlation map, across 50 regions and six weather variables, using PCMCI to find which regions' past conditions carry information about another region's future OLR.

O2. Stress-test every discovered link: does it survive decoy variables, is it physically possible at real atmospheric propagation speeds, does it survive trimming to the strongest links, and does it hold when the regions are redrawn at a different resolution?

O3. Feed the validated graph into the forecaster's causal splitter in three ways (narrowing its candidate sources, nudging its initial scores, or discouraging selections outside the graph) and compare it with a splitter without guidance on unseen years.

O4. Build a drift-detection system that watches each region individually and decides whether to retrain that region's expert, spawn a new one for a changed regime, or reactivate an archived expert that has seen the regime before, avoiding full-model retraining.

Phase 2 (extension).

O5. Design an India-focused hourly forecasting system: a shared graph neural network backbone with region-specific experts conditioned on causally validated drivers, autoregressive forecasts to 72 hours, and station-level validation.

O6. Specify a benchmark against established forecasting models (GraphCast [12] and WeatherNext) and a path to global scale.

1.5 Research Questions and Hypotheses

RQ1

Can PCMCI identify meaningful conditional lagged dependencies among six climate variables across 50 regions?

RQ2

How robust are these dependencies under decoy testing, physical lag checks, top-k selection and a change of spatial resolution?

RQ3

Does the validated graph improve source selection and forecast skill when used as a splitter prior?

RQ4

Does a causal-structure drift signal detect regime shifts earlier than an error signal, and does reusing archived experts pay off?

RQ5

How should these findings shape an India-focused, adaptive forecasting system?

The corresponding testable hypotheses are: (H1) phase-randomised decoys are selected at a rate below 30% and fewer than 5% of links violate physical speed limits; (H2) at least half of each target's outside links persist at double resolution for most targets; (H3) the graph prior improves source discrimination and held-out skill; (H4) the structure channel detects known shifts before the error channel; and (H5) recurring regimes allow archived experts to be reactivated rather than retrained from scratch.

1.6 Scope

Phase 1 forecasts regional-mean OLR anomalies directly at 7-day (primary) and 1-day leads on a 2.5° tropical grid (30°S–30°N, all longitudes) whose BSISO-active core is the Indo-Pacific. Graph discovery uses linear, lagged dependence; contemporaneous links are excluded. All links are statistical statements about conditional dependence between region means, not proofs of physical mechanism. Phase 2 is specified as a design; global forecasting is future work.

1.7 Contributions

1. A **build–validate–freeze–use** protocol in which the causal graph is built and validated on 1979–2012 only and frozen before any test year is used.
2. A **vectorised PCMCI** that matches the Tigramite reference and processes 18,000 candidates per target for 50 targets in about 23 minutes.

3. A **validated causal graph** for BSISO-season OLR: 536 links, 0 of 250 decoys selected, 3 links physically implausible, and 62% mean persistence at double resolution.
4. A **graph-guided splitter** whose prior makes full-pool source scores informative (0 → 24 regions at 7 days, 0 → 36 at 1 day) and improves 1-day skill in 32 of 50 regions.
5. A **dual-channel drift detector and expert lifecycle**: the causal-structure channel detects known shifts several times sooner than the error channel at two of three regions, and 104 of 171 regime changes reuse an archived expert.
6. A **Phase 2 design** that transfers these components to India-focused hourly forecasting.

1.8 System Overview and Thesis Organization

Figure 1.1 summarises the two phases. Chapter 2 reviews the related literature and states the research gap. Chapter 3 defines the data and the forecasting task. Chapter 4 describes the overall Phase 1 workflow and PCMCI. Chapter 5 presents the validation checks, the graph-guided MoE and the drift-adaptive lifecycle. Chapter 6 gives the implementation and experimental design, Chapter 7 the results, Chapter 8 the Phase 2 extension, and Chapter 9 the conclusions. Appendix A lists the configuration, reproduction steps and open decisions.

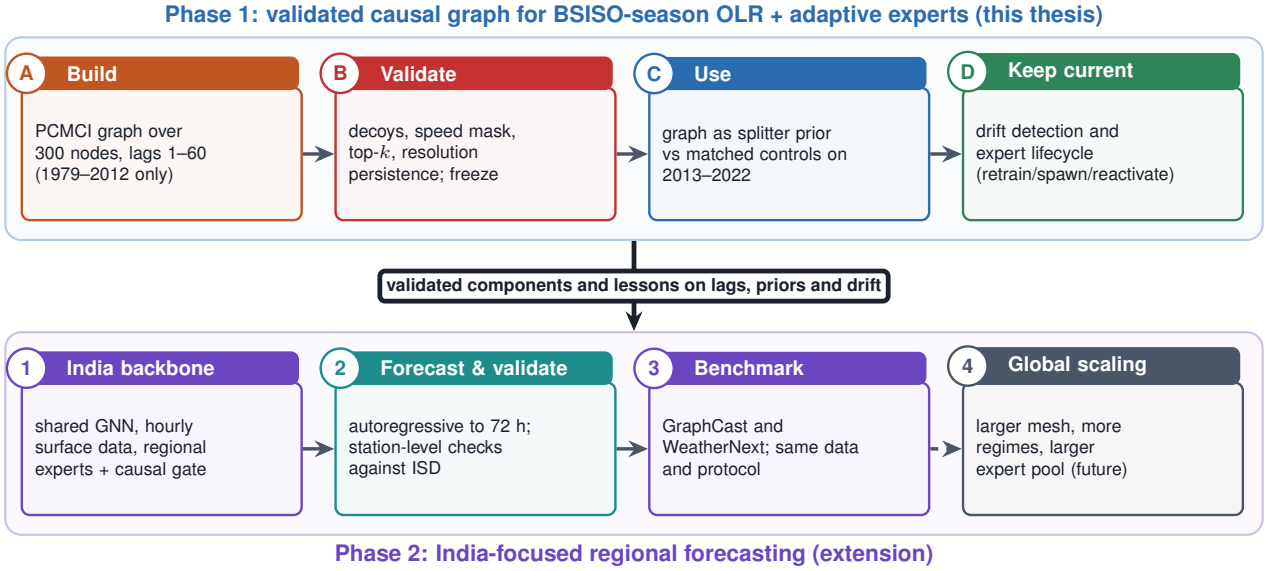


Figure 1.1: Two-phase research roadmap. Phase 1 builds, validates and uses a causal graph and keeps the regional experts current through drift-triggered adaptation. Phase 2 carries the validated components into India-focused regional forecasting, followed by benchmarking and, in future, global scaling.

Chapter 2

Literature Review and Technical Background

2.1 Review Scope

This work draws on four lines of research that are usually studied separately: causal rationale discovery in graph learning, Mixture-of-Experts (MoE) models on graphs, learning under distribution shift, and data-driven weather forecasting. It also builds on causal discovery for climate time series and on tropical intraseasonal dynamics. This chapter reviews each line, compares the closest works in Table 2.1, and states the research gap.

2.2 Tropical Intraseasonal Variability and OLR

Organised tropical convection varies on intraseasonal time scales of 30–60 days. Madden and Julian [6] discovered this variability as an eastward-propagating oscillation in the tropical winds, now known as the MJO. In boreal summer the dominant mode is the BSISO, whose convective envelope forms over the equatorial Indian Ocean and propagates northward over India and the Bay of Bengal and eastward toward the western Pacific, producing a characteristic tilted rain band. Wang and Xie [7] explained this structure as an equatorial Kelvin–Rossby wave packet modified by the monsoon mean flow. Goswami and Ajaya Mohan [9] linked these oscillations to the active and break spells of the Indian monsoon, which makes them central to subseasonal prediction over India [5]. Lee *et al.* [8] defined real-time BSISO indices from a combined empirical orthogonal function analysis of OLR and 850 hPa zonal wind over the Asian monsoon region; this thesis reproduces their recipe to define the active season.

Embedded in the envelope are convectively coupled equatorial waves. Kelvin waves coupled to convection travel eastward at roughly 15–20 $\mathrm{m\,s^{-1}}$ and faster when dry [13]. Signals can therefore cross thousands of kilometres within one or two days, so a purely geographic cutoff on candidate drivers would remove genuine dependencies. These speed scales underlie the physical lag mask of Chapter 5.

OLR has been observed continuously by satellites since the 1970s [10]. Its relation to

cold-cloud fraction and large-scale convective rainfall [11] makes it the standard observable of tropical convection and one of the two fields of the BSISO index.

2.3 Causal Rationales in Graph Learning

Wu *et al.* [14] proposed Discovering Invariant Rationales (DIR) for graph neural networks. A rationale generator splits each input graph into a causal part and a non-causal part. An intervener then swaps the non-causal part between training instances, creating several interventional "environments" for the same causal part. The model is trained to minimise both the mean and the variance of its risk across these environments: if a prediction stays stable when the non-causal part is replaced, the retained part carries invariant, causal information. DIR was designed for graph classification, for example predicting a property of a molecule. It does not address spatiotemporal forecasting, and it provides no mechanism for adapting the model when the data distribution changes.

The splitter used in this thesis adapts the DIR mechanism to regression on a fixed spatial graph. For a target region on a given day, the candidate source regions form the graph, and the non-causal sources are swapped with the same sources on other days (Section 5.7).

2.4 Mixture of Experts on Graphs

The Mixture of Experts combines several specialised networks through a gating function that assigns input-dependent weights [15]. Sparse gating, which activates only a few experts per input, made the approach scale to very large models [16]. In its general form, a mixture of K experts E_k with a gate g predicts

$$\hat{y}(\mathbf{x}) = \sum_{k=1}^{K} g_k(\mathbf{x})\, E_k(\mathbf{x}), \qquad g(\mathbf{x}) = \mathrm{softmax}\Big(\mathrm{TopK}(\mathbf{W}_g \mathbf{x})\Big), \tag{2.1}$$

where TopK keeps the k largest gate logits and sets the rest to $-\infty$, so that only k experts are evaluated. The gate decides *which* expert answers, and the experts decide *what* the answer is. Graph MoE methods differ mainly in what the gate sees and in how the expert pool changes over time. Three recent graph MoE methods are closest to this work.

Topology-conditioned experts. Ghaffari *et al.* [17] forecast traffic with a small router that blends several frozen, pre-trained graph neural network experts at each node, using the node's topology and recent input. The router is cheap and effective, but the experts are frozen: they are never retrained or spawned, so the method has no response to drift.

Curvature-guided routing. Cao *et al.* [18] route each node to expert sub-networks according to its Ollivier–Ricci curvature, operating across Riemannian spaces. Curvature captures local geometry well, but it is a static property of the graph and carries no notion of drift or causality.

Experts that grow over time. Kong *et al.* [19] proposed a dynamic MoE for incremental graph learning that adds a new expert whenever a new batch of data arrives, with a regularisation term to limit forgetting. Experts are added on a schedule, whether or not the data have actually changed.

2.5 Distribution Shift on Graphs

Out-of-distribution generalisation on graphs studies models whose test distribution differs from the training distribution in features, structure or labels [20]. GraphMETRO [21] addresses complex shifts with a mixture of aligned experts: each expert is trained to be invariant to one hand-constructed type of shift, and a learned gate blends them. The approach is effective when the shift types are known, but the shift components are hand-constructed and static, and the method does not discover shift-relevant structure from the data. In the streaming setting, concept-drift methods detect changes in an error or statistic stream [22], for example with ADWIN [23], and climate science uses sequential tests such as STARS [24] to detect regime shifts. Adapting a model after a shift risks catastrophic forgetting of earlier regimes [25].

2.6 Causal Discovery for Climate Time Series

Granger causality [26] asks whether the past of one series improves the prediction of another. In climate data, lagged correlation and naive Granger tests are distorted by autocorrelation and common drivers [1]. PCMCI [3] addresses this in two stages: a fast condition-selection step finds a small superset of each target's parents, and a momentary conditional-independence (MCI) test then checks each candidate while conditioning on the parents of both the target and the source. PCMCI is the reference method for causal discovery in Earth-system time series [2, 4]; extensions such as LPCMCI handle latent confounders [27]. It has been used to study monsoon teleconnections [28], typically over tens of pre-selected indices rather than hundreds of regional nodes, and the discovered graph is usually an end in itself rather than an input to a forecasting model.

Because thousands of candidate links are tested, multiple-testing control is essential; the Benjamini–Hochberg procedure controls the false discovery rate [29]. Surrogate data, series that keep chosen statistical properties of the real data but destroy their coupling [30, 31], provide a direct test of whether a discovery procedure selects spurious sources.

2.7 Data-Driven Weather Forecasting

GraphCast [12] showed that a graph neural network trained on reanalysis can outperform operational medium-range forecasting on most variables and leads, and limited-area versions extend the idea to regional domains [32]. These models learn dependencies between regions

implicitly. They do not expose a tested statement of which regions drive a target, and they do not adapt when those dependencies change.

2.8 Consolidated Literature Review

Table 2.1: Consolidated literature review: key idea and limitation of the closest related works.

Paper	**Key idea / method**	**Limitation**
Wu *et al.*, DIR, ICLR 2022 [14]	Separates graph information into causal rationale and non-causal structure; an intervener swaps the non-causal part across instances; stable prediction across interventions indicates invariant causal information.	Designed for graph classification; does not address spatiotemporal weather forecasting; no dynamic expert adaptation.
Ghaffari *et al.*, GC-MoE, 2026 [17]	A small router blends several frozen, pre-trained experts per node based on topology and recent input.	Experts are frozen: no retraining, no spawning, no response to drift.
Cao *et al.*, GeoMoE, 2026 [18]	Routes each node to expert sub-networks based on Ollivier–Ricci graph curvature across Riemannian spaces.	Routes by topology and curvature alone; no notion of drift or causality.
Wu *et al.*, GraphMETRO, NeurIPS 2024 [21]	Experts are each trained to be invariant to one hand-constructed shift type and blended by a learned gate.	Shift components are hand-constructed and static; does not discover shift-relevant structure from data.
Kong *et al.*, DyMoE, 2025 [19]	Spawns a new expert for each incoming data batch, with regularisation against forgetting.	Experts are added on a fixed schedule, not because a change is detected; no causal structure.
Runge *et al.*, PCMCI, 2019 [3]	Condition selection followed by momentary conditional-independence tests for high-dimensional autocorrelated time series.	Produces a graph but is not connected to a forecasting model or to expert adaptation.

Paper	Key idea / method	Limitation
Lam *et al.*, GraphCast, 2023 [12]	Graph neural network trained on reanalysis for skilful medium-range global forecasting.	Dependencies are implicit; no causal source selection and no drift adaptation.
This thesis	Validated PCMCI graph as the splitter prior of a regional MoE; dual-channel (error + causal-structure) drift detection; experts retrained, spawned or reactivated per region.	Phase 1 on daily OLR at 50 regions; India-focused hourly system specified as Phase 2.

2.9 Research Gap

Existing studies address causal rationale discovery, graph distribution shifts, dynamic expert learning and graph-based weather forecasting separately. They do not combine causal structure discovery with drift-triggered expert adaptation for spatiotemporal weather forecasting. Specifically:

G1. Causal source selection is not validated. Causal rationales in DIR are learned end-to-end and never checked independently; causal graphs from PCMCI are checked but not used to guide a forecaster.

G2. Experts do not respond to measured drift. Graph MoE methods route by topology or curvature (GC-MoE, GeoMoE), keep experts frozen (GC-MoE), use hand-constructed shift types (GraphMETRO), or add experts on a fixed schedule (DyMoE).

G3. Drift is not detected through causal structure. Concept-drift detectors watch error streams, which react only after forecasts have already degraded.

This thesis addresses the gap by validating causal drivers first, using them to guide regional experts, and adapting only the regions whose behaviour has measurably changed.

Chapter 3

Problem Formulation, Dataset, and Preprocessing

3.1 Forecasting Task

Let Y_t^r be the standardised OLR anomaly of region $r \in \{0, \ldots, 49\}$ on day t. For lead L days, a model for region r predicts

$$\hat{Y}_{t+L}^r = f_r\Big(\{X_{t-\ell}^i : i \in \mathcal{N}_r,\ \ell \in \Lambda\}\Big), \tag{3.1}$$

where X_t^i is the standardised anomaly of node i, $\mathcal{N}_r$ the nodes of the source regions selected for region r, and Λ the input lags. Forecasts are *direct* (one model per lead, no roll-out); the primary lead is $L = 7$ days and the secondary lead $L = 1$ day. The central question is how $\mathcal{N}_r$ should be chosen. A forecast is scored only if its target day lies in the BSISO season $\mathcal{S}$ and in the test period 2013–2022; inputs may come from any earlier day.

3.2 Dataset

The data are BSISO monitoring fields on a 2.5° grid (25 latitudes from 30°N to 30°S × 144 longitudes, i.e. 3,600 cells of about 275 km), daily from 1 January 1979 to 31 December 2022: 16,071 days with none missing. Each field was anomaly-processed upstream, with the seasonal cycle of each location and calendar day removed (climatology base period 1979–2014) and the anomalies normalised per grid cell. Table 3.1 lists the six variables.

3.3 Regions and the 300-Node Representation

Single grid cells are too noisy and too numerous for a 60-lag causal analysis, so the cells are grouped into 50 regions by k-means clustering of their OLR time series over the development period; 46 of the 50 regions are contiguous, with about 72 cells each. A region adjacency graph, built from which regions touch on the map with longitude wrap-around, defines the local

Table 3.1: The six variables and their role in tropical convection. All are used as standardised anomalies.

Field	Variable	Role
SST	Sea surface temperature	Slow surface boundary condition; modulates the heat and moisture fluxes that sustain convection
h850	Geopotential height, 850 hPa	Low-level pressure and wave response to convective heating
u200	Zonal wind, 200 hPa	Upper-level outflow and large-scale overturning above convection
PW	Precipitable water	Column moisture; a strong precursor of deep convection
u850	Zonal wind, 850 hPa	Low-level monsoon flow and moisture convergence; a BSISO index field
OLR	Outgoing longwave radiation	Convection proxy (low OLR = active convection); the forecast target and a BSISO index field

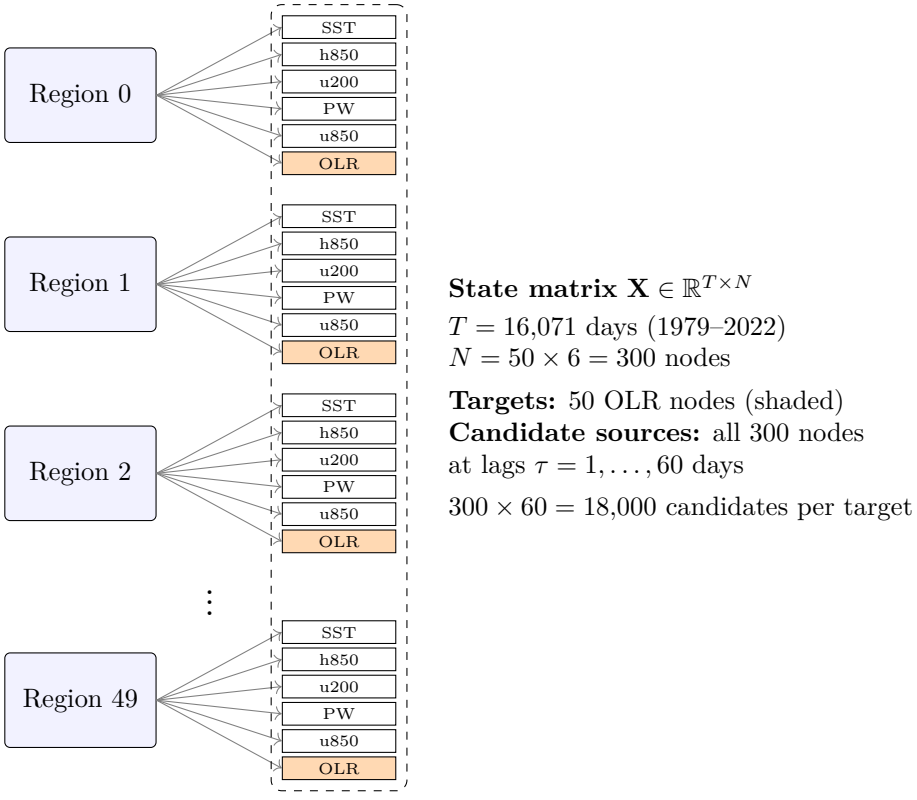


Figure 3.1: Schematic of the 300-node representation: each of the 50 regions contributes six regional-mean variables; the 50 OLR nodes are the targets and all 300 nodes are candidate sources. Schematic only; real region shapes are shown in Figure 7.2.

candidate pools of the forecaster. Each node $i = (r, v)$ is the mean of variable v over region r (SST over ocean cells only), re-standardised with development-period statistics:

$$X_t^i = \frac{Z_t^i - \mu_i}{\sigma_i}, \qquad \mu_i, \sigma_i \text{ computed on 1979–2012.} \tag{3.2}$$

With 50 regions and six variables there are $N = 300$ nodes (Figure 3.1). For each OLR target, every node at every lag from 1 to 60 days is a candidate source: $300 \times 60 = 18{,}000$ candidates per target and 900,000 for all 50 targets. For the resolution check, each region is split into two by k-means, giving 100 regions and 596 nodes (four sub-regions have no ocean cells and therefore no SST node).

The forecaster describes each region on each day by a 49-value feature vector: the six variables at lags $\Lambda = \{0, 1, 2, 3, 5, 7, 10\}$ days, plus the region's ocean fraction.

3.4 BSISO Season

The BSISO is a summer phenomenon, so target days are restricted to its active season, defined from the oscillation's own amplitude rather than a fixed calendar window. The index follows Lee *et al.* [8]: OLR and u850 anomalies over 10°S–30°N, 40°–160°E (the grid ends at 30°N), with the preceding 120-day mean removed, $\sqrt{\cos\varphi}$ area weighting, and a combined EOF fitted on May–October days of 1979–2012 only. The amplitude is $A_t = \sqrt{\mathrm{PC1}_t^2 + \mathrm{PC2}_t^2}$. In each year, the season runs from the first to the last stretch of at least 20 days in which the 31-day mean amplitude exceeds 1, padded by 21 days at each end. The season averages 208 days per year (range 82–286), giving 7,153 discovery days (1979–2012) and 1,989 scored test days (2013–2022).

The season restricts *target* days only. A source at lag 45 for a target on 1 June is read from mid-April; deleting non-season days before building lags would remove precisely the precursor information under study.

3.5 Development and Test Split and Leakage Control

Everything that shapes the graph or the models uses 1979–2012 only (Figure 3.2, Table 3.2): node standardisation, the regional partition, the BSISO EOF, PCMCI and its validation, and the choice of k. The forecaster trains on 1979–2004 and validates on 2005–2012. The graph is frozen before 2013–2022 is used, and the test years are used only for scoring. A chronological split is used because random splits of autocorrelated series place near-duplicate days in both training and test sets. The upstream anomaly climatology (1979–2014) overlaps the test period by two years; all compared models share it, and refitting it on 1979–2012 is listed as future work.

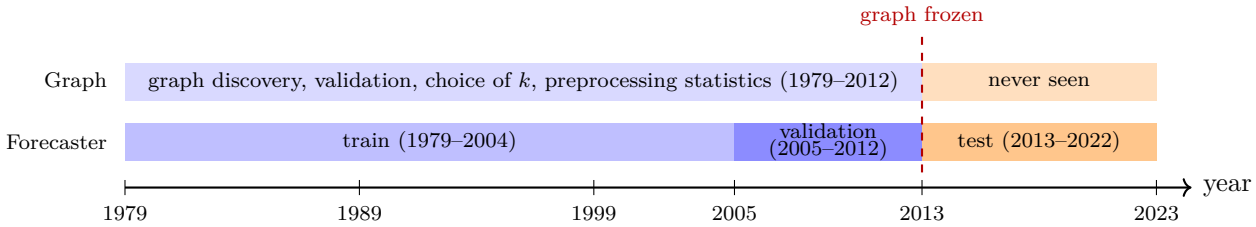


Figure 3.2: Phase 1 time split. Everything that shapes the graph or the forecaster uses 1979–2012 only; the graph is frozen before 2013–2022, which is used only for scoring.

Table 3.2: Leakage channels and controls.

Channel	**Control**
Node standardisation, regional partition	Statistics and clustering fitted on 1979–2012
BSISO EOF patterns	Fitted on May–October days of 1979–2012
Graph discovery and validation	PCMCI, decoys, speed mask and persistence on 1979–2012 season days
Choice of k	Fit 1979–2004, score 2005–2012
Forecaster	Train 1979–2004, validate 2005–2012; test years used only for scoring
Season mask	Selects which target days are scored; never an input to a forecast

3.6 Chapter Summary

The data are six daily variables on a tropical grid, grouped into 50 OLR-coherent regions and 300 nodes. Targets are BSISO-season OLR anomalies at 7-day and 1-day leads, scored on held-out 2013–2022 days, with all modelling choices made on 1979–2012.

Chapter 4

Proposed Methodology: PCMCI and Graph Discovery

4.1 Phase 1 Workflow

Figure 4.1 shows the complete Phase 1 workflow. The data (step 1) are reduced to 50 regions and 300 nodes (step 2), and the BSISO season and time split are fixed (step 3). Stage A discovers a lagged causal graph with PCMCI (step 4); Stage B validates it with four independent checks (step 5); and the graph is merged into a region-level prior and frozen (step 6). Stage C uses the prior to guide a causal splitter and per-region expert (step 7), which produce the OLR forecast (step 8). Forecast errors and causal-structure changes feed a dual-channel drift detector (step 9) that decides when a region's expert is retrained, archived, reactivated or replaced (step 10). Stage D tracks whether the causal structure itself changes over time (step 11). This chapter describes Stage A; Chapter 5 describes Stages B–D and the expert lifecycle.

4.2 Why Correlation Is Not Enough

Three mechanisms make a source correlated with a target without influencing it directly. *Autocorrelation*: a source that truly acts at lag 2 also appears correlated at lags 1, 3 and 4 because it is persistent. *Common drivers*: an active BSISO phase changes winds, moisture and OLR over large areas at once, so two regions appear linked through their shared driver. *Mediation*: a source acting through an intermediate region appears linked to the target directly. In each case the spurious dependence disappears once the right variables are conditioned on. Conditional independence, written $X \perp\!\!\!\perp Y \mid Z$, formalises this idea, and causal discovery tests it systematically [2].

4.3 Partial-Correlation Test

The conditional-independence test is partial correlation (ParCorr). Given samples of x, y and a conditioning matrix $\mathbf{Z}$ with orthonormal basis $\mathbf{Q}$, the residuals $\mathbf{r}_x = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{x}$ and

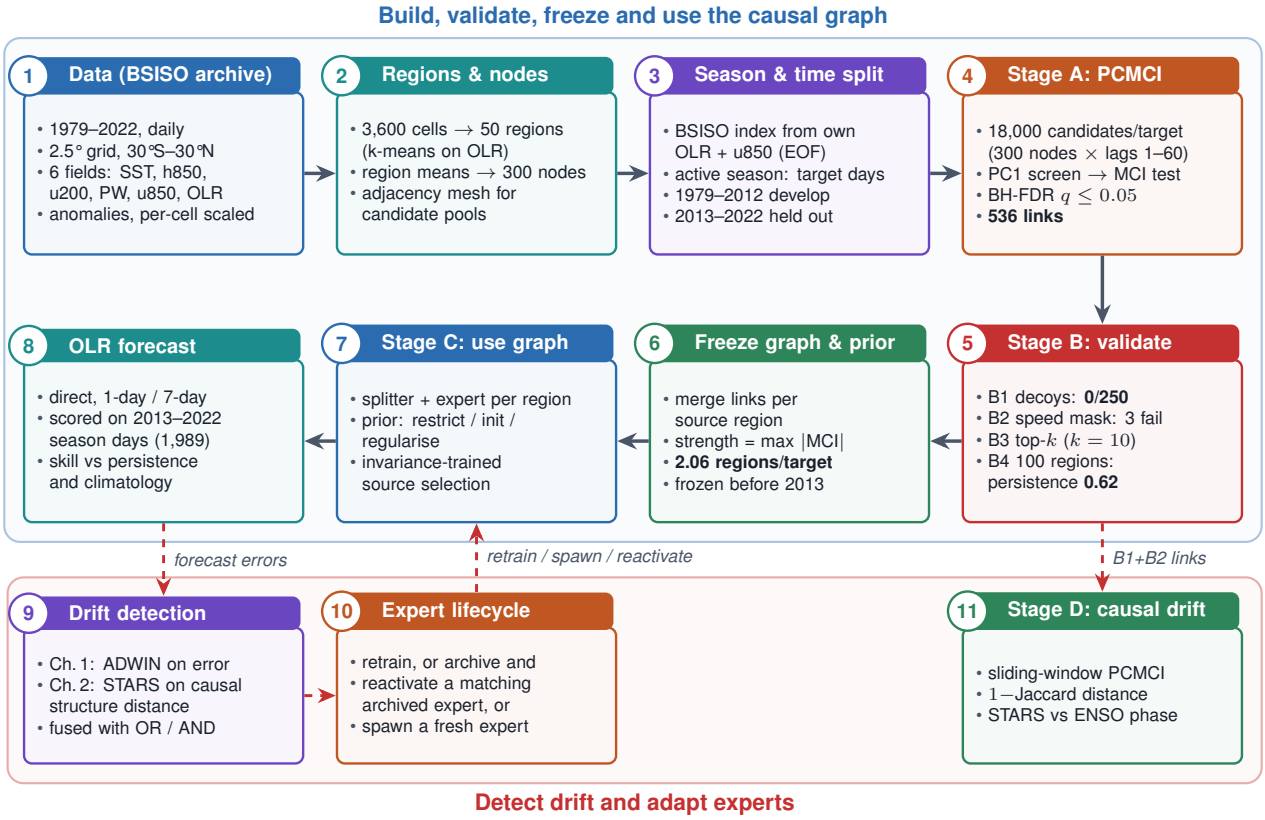


Figure 4.1: Phase 1 workflow. Steps 1–8 build, validate, freeze and use the causal graph; steps 9–11 detect drift and adapt the regional experts. Numbers in bold are measured results reported in Chapter 7.

$\mathbf{r}_y = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{y}$ give

$$\hat{\rho}(x, y \mid \mathbf{Z}) = \frac{\mathbf{r}_x^\top \mathbf{r}_y}{\|\mathbf{r}_x\| \, \|\mathbf{r}_y\|}, \qquad t = \hat{\rho}\sqrt{\frac{n - 2 - |\mathbf{Z}|}{1 - \hat{\rho}^2}}, \tag{4.1}$$

where t follows a Student-t distribution with $n - 2 - |\mathbf{Z}|$ degrees of freedom under the null hypothesis $\rho = 0$. ParCorr detects linear dependence.

4.4 Stage 1: PC1 Condition Selection

For each OLR target X_t^j, all 18,000 candidates $X_{t-\tau}^i$ ($\tau = 1, \ldots, 60$) start in the parent set. At iteration ℓ, each candidate is tested for $X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \mathbf{S}_\ell$, where $\mathbf{S}_\ell$ holds the ℓ strongest other candidates found so far, and it is dropped if the p-value exceeds $\alpha_{\mathrm{PC}} = 0.05$. Iterations continue while enough candidates remain. Because the strongest drivers, typically the target's own recent past, are conditioned on first, most autocorrelation-induced and indirect candidates are removed in the first few rounds. The result is a small superset $\hat{\mathcal{P}}(X_t^j)$ of the true parents. Non-target nodes are screened only against their own past, which gives the autocorrelation structure needed in the next stage.

4.5 Stage 2: Momentary Conditional Independence

Each survivor is retested while conditioning on the target's other parents *and* on the source's own parents, shifted by the lag:

$$\hat{\rho}_{i \to j}^{\mathrm{MCI}}(\tau) = \hat{\rho}\Big(X_{t-\tau}^i,\, X_t^j \;\Big|\; \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\},\, \hat{\mathcal{P}}(X_{t-\tau}^i)\Big). \tag{4.2}$$

Conditioning on the target's parents removes indirect paths and common drivers among them. Conditioning on the source's parents removes the source's own memory, so the test measures the information the source adds at lag τ beyond its history. This calibrates the test for strongly autocorrelated climate data and sharpens the lag estimate [3]. The value $\hat{\rho}^{\mathrm{MCI}}$ is used as the link strength.

A worked example. Suppose region A's OLR drives region B's OLR at lag 2, A is strongly persistent, and an active BSISO phase raises the moisture of both regions simultaneously. The lagged correlation between A and B is then significant at lags 1–5, and region A's moisture also appears correlated with B. PC1 conditions on B's own recent past and on A at lag 2, the strongest candidate, which removes most of the neighbouring lags. MCI then conditions A at lag 3 on A's own past at lags 4 and beyond: the remaining dependence at lag 3 is merely A's persistence, so the test is not significant. Conditioning on B's moisture parent removes the shared moisture signal. Only the direct link at lag 2 survives, with a strength that measures A's unique contribution.

4.6 False-Discovery-Rate Control

Testing 18,000 candidates per target at $\alpha = 0.05$ would admit about 900 false links per target even if none existed. The Benjamini–Hochberg procedure [29] controls the expected fraction of false links among those selected. With $m = 18{,}000$ sorted p-values $p_{(1)} \le \cdots \le p_{(m)}$,

$$q_{(h)} = \min_{h' \ge h} \min\Big(1,\ \frac{m}{h'}\, p_{(h')}\Big), \tag{4.3}$$

and a link is retained if $q \le 0.05$. Candidates dropped by PC1 count with $p = 1$, so the correction covers all 18,000 hypotheses. Lag 0 is excluded, so every link points forward in time.

4.7 Vectorised Implementation

The reference Tigramite implementation builds a new data array for every test, which is too slow at 18,000 candidates per target. Our implementation (Algorithm 4.1) builds the lagged design matrix of all candidates once, shares it across the 50 targets, and tests all candidates that share a conditioning set in one batched projection, a single QR decomposition followed by one matrix product. It was verified against Tigramite on synthetic data with known parents, including a common driver and a noise node: the PC1 parent sets are identical, and the MCI values agree to within 2×10^{-3}.

Algorithm 4.1 Vectorised PCMCI for one OLR target

Require: node series $\mathbf{X}$; season days $t \in \mathcal{S}$, 1979–2012; target j; $\tau \in [1, 60]$; $\alpha_{\mathrm{PC}} = \alpha_{\mathrm{FDR}} = 0.05$

Build design $\mathbf{D}$ with one column per candidate (i, τ) ▷ shared by all targets
$C \leftarrow$ all 18,000 candidates; $\ell \leftarrow 0$
while $|C| - 1 \ge \ell$ **do** ▷ PC1
    $\mathbf{S} \leftarrow$ the ℓ strongest members of C; orthonormalise $\mathbf{D_S}$ by QR
    residualise the target and all of $\mathbf{D}_C$ in one product; compute $\hat{\rho}$ and p-values (Eq. 4.1)
    drop candidates with $p > \alpha_{\mathrm{PC}}$; $\ell \leftarrow \ell + 1$
end while
for all $(i, \tau) \in C$ **do** ▷ MCI
    test $X^i_{t-\tau} \perp\!\!\!\perp X^j_t$ given the target's other parents and the source's lagged parents (Eq. 4.2)
end for
compute q over all 18,000 candidates (Eq. 4.3) and keep links with $q \le \alpha_{\mathrm{FDR}}$

4.8 Assumptions

PCMCI identifies causal parents under causal sufficiency (all common drivers are observed), stationarity, and faithfulness. In regional climate data these hold only approximately: the BSISO phase, ENSO and sub-regional moisture processes are only partly represented by 300 region means; dependence varies between years; and convection responds nonlinearly to moisture. The fields within one region also move together, so which field PCMCI selects can vary even when the source region is stable. For these reasons the graph is treated as a hypothesis to be

validated (Chapter 5), and claims are made at the level of source regions. Methods that admit latent confounders, such as LPCMCI [27], are a natural extension.

Chapter 5

Graph Validation and the Graph-Guided Mixture of Experts

5.1 Why the Graph Needs Independent Validation

There is no ground truth for causal links in the real atmosphere, and PCMCI's guarantees rest on assumptions that hold only approximately. Stage B therefore applies four checks in a fixed order, each of which can *reject* links even though none can prove a link correct (Figure 5.1). The resulting graph is frozen before any test year is used.

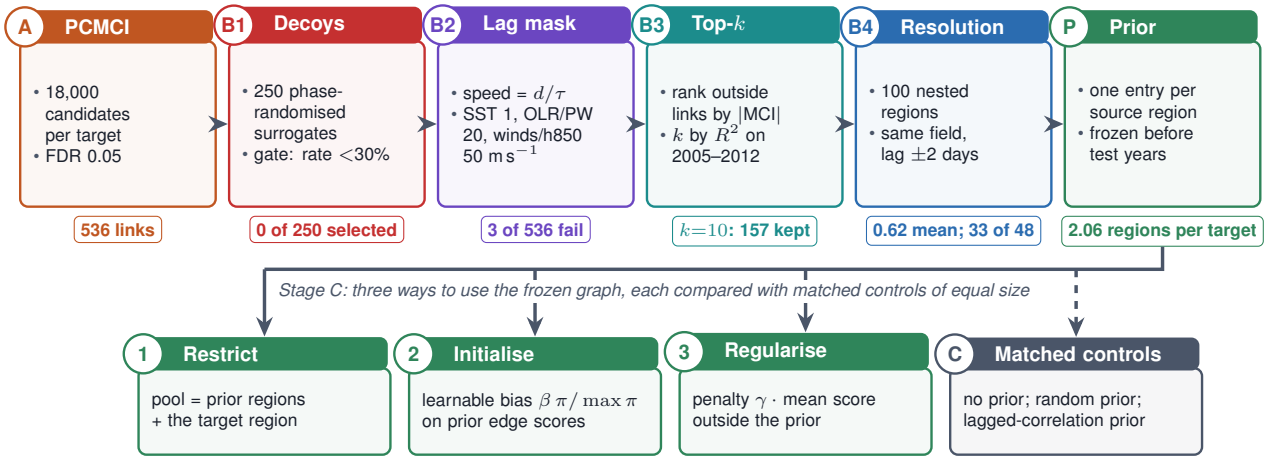


Figure 5.1: Stage B validation and Stage C use of the graph. The tags under each check give the measured outcome (Chapter 7). The frozen region-level prior enters the splitter in one of three modes and is compared with matched controls of equal size.

5.2 B1: Surrogate Decoys

The first check asks whether the pipeline selects series merely because they have realistic memory. Decoys are phase-randomised copies of real nodes [30, 31]. A series x_t is Fourier-transformed,

its phases are replaced by independent uniform random phases, and it is transformed back:

$$\tilde{x}_t = \frac{1}{n} \sum_f |\hat{x}_f| \, e^{i\theta_f} \, e^{2\pi i f t/n}, \qquad \theta_f \sim \mathcal{U}[0, 2\pi). \tag{5.1}$$

The decoy keeps the power spectrum, and therefore the autocorrelation including the 30–60-day BSISO peak, but has no real relationship with any node. 250 decoys are added to the 300 real nodes in a separate PCMCI pass (550 nodes, 33,000 candidates per target). The *decoy rate* δ is the fraction of decoys selected as a parent of any target, and the pre-set warning line is $\delta \geq 30\%$. Decoys do not reproduce non-Gaussian amplitudes, seasonal changes in variance, or a common driver shared with the target, so a low decoy rate rules out autocorrelation-driven selection but does not by itself prove causality.

5.3 B2: Physical Lag Mask

A link from region r' to region r at lag τ requires a signal to travel at least the closest great-circle distance $d(r', r)$ between the two regions in τ days. Its implied speed is

$$s_e = \frac{d(r', r)}{\tau \times 86{,}400\,\mathrm{s}}, \tag{5.2}$$

and the link fails if s_e exceeds a generous limit for its source field: $1\,\mathrm{m\,s^{-1}}$ for SST, whose own influence is ocean-mediated; $20\,\mathrm{m\,s^{-1}}$ for OLR and PW, the upper range of convectively coupled Kelvin waves [13]; and $50\,\mathrm{m\,s^{-1}}$ for the wind and height fields, which can be carried by fast equatorial waves. Unlike a distance cutoff, this test keeps remote teleconnections that act at plausible speeds. Links at lags of 25 days or more are flagged as possible echoes of the 30–60-day oscillation. Failing links are removed from the forecasting prior but kept, with their flags, in the graph file for inspection.

5.4 B3: Data-Driven Top-k

Each target's outside links, those from other regions, are ranked by $|\hat{\rho}^{\,\mathrm{MCI}}|$. For each $k = 0, \ldots, 10$, a ridge regression on the target's own-region links plus its top-k outside links is fitted on 1979–2004 season days and scored by R^2 on 2005–2012; k is chosen as the maximiser. The test period plays no part. The step sets the size of the prior from data and caps the number of drivers per target.

5.5 B4: Resolution Persistence

A link that is an artefact of how the regions were drawn should be fragile under a different partition. Stage A is rerun at 100 nested regions (each region split in two). An outside link persists if some child pair carries a link from the same field at a lag within ± 2 days. The

persistence score of target r is the fraction of its outside links that persist, and a target passes if $\Pi_r \geq 0.5$. Failures are flagged rather than deleted, because a real large-scale signal can be diluted at a finer scale.

5.6 From Validated Links to a Region-Level Prior

The splitter chooses whole source regions, and the field chosen within a region is the least stable part of the graph. Validated outside links are therefore merged per source region:

$$\pi_r(r') = \max_{e:\ \mathrm{src}(e)=r'} |\hat{\rho}_e^{\mathrm{MCI}}|, \tag{5.3}$$

over the links that pass B1 and B2 and survive B3. Field and lag details remain in the frozen graph file.

5.7 The Graph-Guided Mixture of Experts

Each region has its own causal splitter and expert, 50 independent pairs trained from scratch with no weight sharing. Over time each region keeps a lineage of experts managed by the lifecycle of Section 5.9, which makes the system a Mixture of Experts in which a region's active expert can change with the regime [15].

Candidate pools. The splitter of target r considers either its adjacent regions (*direct*, about 6–8), neighbours of neighbours (*2hop*, about 15–23), or all 50 regions (*full*). The target region is always a candidate.

Causal splitter. Following DIR [14], a rationale generator scores each candidate edge from a 10-day window of the two regions' histories,

$$a_{r'r,t} = \sigma\Big(\mathrm{MLP}[\mathbf{w}_t^{r'} \,\|\, \mathbf{w}_t^{r}] + b_{r'r}\Big), \tag{5.4}$$

where $b_{r'r}$ is an optional prior bias. The top fraction $\min(0.5, 3/|\mathcal{C}_r|)$ of edges forms the causal set; the remaining edges are swapped with the same edges on $M = 6$ other days to create interventions. The splitter minimises the mean and variance of the risk across interventions,

$$\mathcal{L}_{\mathrm{split}} = \frac{1}{M}\sum_m \ell_{t,m} + \lambda\,\mathrm{Var}_m[\ell_{t,m}], \tag{5.5}$$

with λ warmed up from 0 to 1. A spurious head with detached gradients serves as a leakage check.

Expert. The expert aggregates the selected sources' 49-value feature vectors, weighted by their splitter scores, and maps the result to the forecast with a small MLP. The target's own

history enters only if the target region is selected, so the forecast depends on the splitter's choice. A splitter is *informative* if the spread of its candidates' mean scores is $\sigma_a \geq 0.02$; below that the scores cluster around one value and the selection is effectively random.

5.8 Three Ways to Use the Graph

- **Restrict:** the candidate pool becomes the prior regions plus the target, $\mathcal{C}_r = \{r' \in \pi_r\} \cup \{r\}$.
- **Initialise:** the pool is unchanged, and the learnable bias in Eq. (5.4) starts at $b_{r'r} = 2\,\pi_r(r')/\max \pi_r$ for prior regions and 0 otherwise.
- **Regularise:** the pool is unchanged, and a penalty $\gamma \cdot \mathrm{mean}(a_{r'r})$ over candidates outside the prior is added to the loss.

To attribute an effect to the graph's content, each mode is compared with matched controls of equal size per target: no prior, a random prior (the same number of regions drawn at random with the strengths shuffled), and a lagged-correlation prior (the regions most correlated with the target on training data).

A link at lag τ can inform a direct forecast at lead L only if $\tau \geq L$, because the source value it describes must be known at the issue time. A graph whose links act within one or two days is therefore matched to short-lead forecasts.

5.9 Drift Detection and the Expert Lifecycle

Figure 5.2 shows how each region's expert is kept current.

Channel 1: error (ADWIN). ADWIN [23] keeps an adaptive window of recent forecast errors. Whenever two sub-windows have significantly different mean errors, it drops the older part and raises an alarm. For a window W split into sub-windows W_0 and W_1 with means $\hat{\mu}_0, \hat{\mu}_1$ and sizes n_0, n_1, a change is declared if

$$|\hat{\mu}_0 - \hat{\mu}_1| > \epsilon_{\text{cut}} = \sqrt{\frac{1}{2m} \ln \frac{4|W|}{\delta}}, \qquad m = \left(\frac{1}{n_0} + \frac{1}{n_1}\right)^{-1}, \tag{5.6}$$

where δ is the confidence parameter.

Channel 2: causal structure (STARS). The splitter's per-edge score vector on each day is the region's causal signature; its cosine distance from a reference signature (the mean over a baseline window) forms a scalar causal-structure series. Rodionov's STARS test [24] flags a point that falls outside the current regime's range and confirms a shift only if the following ℓ

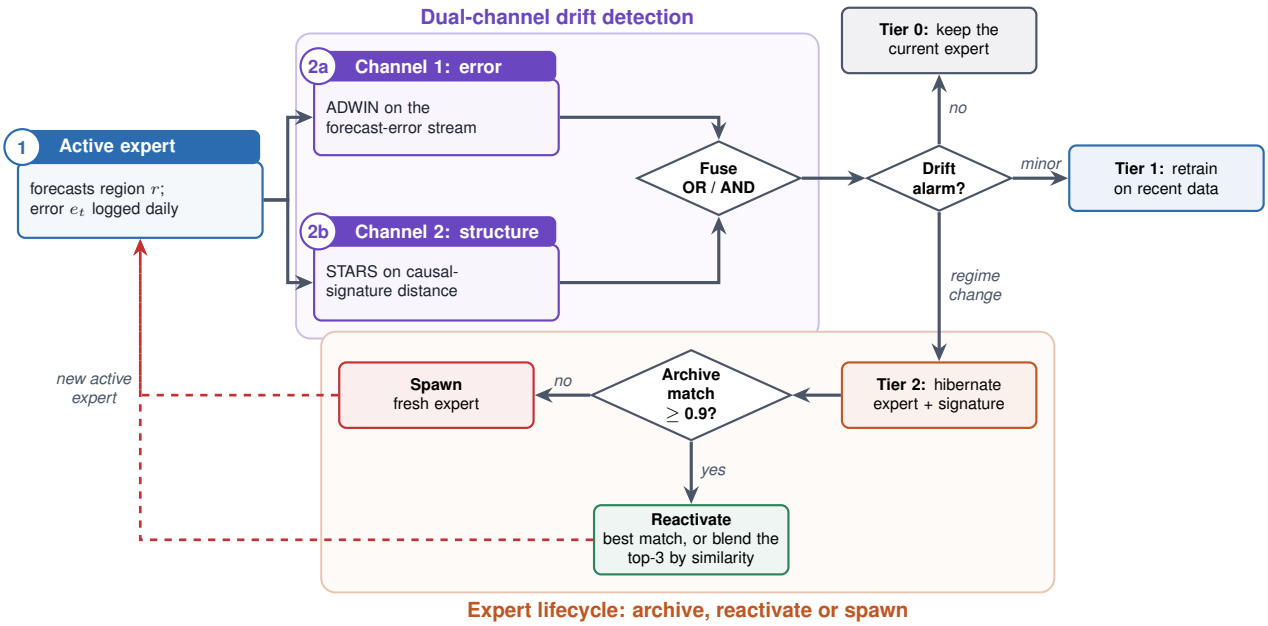


Figure 5.2: Drift detection and the expert lifecycle for one region. Two channels watch the forecast error and the causal structure; a fused alarm leads to retraining, or to archiving the expert and then reactivating a matching archived expert or spawning a fresh one.

points support the new level. For a candidate shift at time j, the regime-shift index

$$\mathrm{RSI}_j = \sum_{i=j}^{j+m} \frac{x_i - \bar{x}_{\mathrm{crit}}}{\ell\,\hat{\sigma}_\ell}, \qquad m = 0, \ldots, \ell - 1, \tag{5.7}$$

must keep its sign over the cut-off length ℓ, where $\bar{x}_{\mathrm{crit}}$ is the level that a new regime's mean must exceed at the chosen significance and $\hat{\sigma}_\ell$ is the average standard deviation over intervals of length ℓ. Because it watches the structure rather than the outcome, it can in principle detect a change before forecasts degrade.

Fusion. The channels are combined with OR, which reacts faster but raises more alarms, or with AND, which is slower but has fewer false alarms.

Lifecycle. After an alarm, a minor change triggers retraining (Tier 1). A regime change (Tier 2) hibernates the current expert together with its causal signature and searches the archive. If an archived expert's signature has cosine similarity of at least 0.9, it is reactivated, either the single best match or a similarity-weighted blend of the three best matches; otherwise a fresh expert is spawned. Because climate regimes such as ENSO phases recur, reactivation reuses knowledge instead of relearning it, and adding experts rather than overwriting them limits catastrophic forgetting [25].

Stage D: structural drift of the PCMCI graph. Separately, PCMCI is rerun on sliding windows of one or three BSISO seasons, restricted to each target's B1/B2-validated candidates. The distance $\Delta_w = 1 - J(\mathcal{P}_w, \mathcal{P}_{w-1})$ between consecutive parent sets, where J is the Jaccard

similarity, is tested with STARS and compared with ENSO phase changes derived from the project's own SST.

5.10 Consolidated Algorithm

Algorithm 5.1 summarises Phase 1. Every step before the freeze uses 1979–2012 only.

Algorithm 5.1 Build, validate, freeze and use the causal graph, and keep experts current

compute the BSISO index and season $\mathcal{S}$ (EOF fitted on 1979–2012)
Stage A: run PCMCI (Algorithm 4.1) for each of the 50 OLR targets
B1: add 250 phase-randomised decoys in a separate pass; stop if the decoy rate is $\geq 30\%$
B2: compute implied speeds; remove links above the field's limit from the prior
B3: choose k by validation R^2; keep the top-k outside links per target
B4: rerun PCMCI at 100 regions; flag targets with $\Pi_r < 0.5$
build the region prior π_r (Eq. 5.3) and **freeze** graph and prior
Stage C: train a splitter and expert per region with the prior (restrict, initialise or regularise) and with matched controls; score on 2013–2022
for each day in the monitored period **do** ▷ lifecycle
    update ADWIN (error) and STARS (structure); fuse with OR or AND
    if alarm **then** retrain, or hibernate and reactivate a matching archived expert (similarity ≥ 0.9), or spawn
    end if
end for
Stage D: sliding-window PCMCI, Δ_w, STARS and comparison with ENSO

Chapter 6

Implementation and Experimental Design

6.1 Implementation

The system is a Python library (`causal_moe`) with one script per experiment and an automated test suite, run on CPU. NumPy and SciPy implement the vectorised PCMCI and the validation checks, scikit-learn the regional clustering, PyTorch and PyTorch Geometric the splitter and experts, and the `river` library the ADWIN detector; STARS is implemented from its published algorithm. The PCMCI tests compare our implementation with Tigramite (Section 4.7). Every experiment writes its configuration and per-region results to JSON, and all figures are regenerated from these files.

Table 6.1: Software environment (versions pinned in the project's requirements file).

Component	Version	Use
NumPy / SciPy	1.26.4 / 1.17.1	Vectorised PCMCI, FFT surrogates, statistics
scikit-learn	1.9.0	k-means regional partition
PyTorch (CPU) / PyTorch Geometric	2.13.0 / 2.8.0	Splitter and experts
river	0.22.0	ADWIN drift detector
Tigramite	5.2.10.1	Reference PCMCI for verification
pytest	9.0.3	Automated tests

6.2 Configuration

6.3 Experiments

Table 6.3 lists the experiments reported in Chapter 7. Stage C compares the PCMCI prior in restrict mode with the no-prior splitter in the full 50-region pool, the setting in which the splitter alone cannot discriminate and a prior has most room to help. The complete design crosses three priors (PCMCI, random, correlation) with three modes and three pools; the remaining cells use the same, already tested code and are listed as future work.

Table 6.2: Configuration of the causal graph (Stages A–B) and of the splitter and expert (Stage C). Splitter and expert settings are identical for every compared condition.

Parameter	**Value**
Stage A: PCMCI	
Nodes; candidates per target	300 (50 regions × 6 fields); 18,000
Lags; test	1–60 days (lag 0 excluded); ParCorr
α_{PC}; FDR level	0.05; 0.05 (Benjamini–Hochberg over all candidates)
Discovery days	7,153 BSISO-season days, 1979–2012
Stage B: validation	
B1 decoys; warning line	250 phase-randomised; decoy rate $\geq 30\%$
B2 speed limits	SST 1, OLR/PW 20, h850/u850/u200 50 $\mathrm{m\,s^{-1}}$; flag lags ≥ 25 d
B3	$k \in \{0, \ldots, 10\}$; fit 1979–2004, R^2 on 2005–2012
B4	100 regions (596 nodes); same field, lag ± 2 d; pass $\Pi_r \geq 0.5$
Stage C: splitter and expert (one pair per region)	
Epochs; training samples	5; 4,000 contiguous samples
Optimiser; learning rate; batch	Adam; 10^{-3}; 16
Hidden width; generator window	16; 10 days
Interventions M; λ	6; 0 for the first 30% of steps, then ramped to 1
Selected fraction	$\min(0.5,\ 3/\lvert\mathcal{C}_r\rvert)$
Train / validation / test	1979–2004 / 2005–2012 / 2013–2022 (1,989 season days)
Lifecycle	
Archive match threshold; pool blend	similarity ≥ 0.9; top-3 similarity-weighted

Table 6.3: Experiments reported in this thesis.

Experiment	**Setting**	**RQ**
Stage A + B	PCMCI at 50 regions; B1–B4; frozen prior	1, 2
Pool size baseline	No prior, direct / 2hop / full, 7 days, year-round and season	3
Stage C	PCMCI restrict vs no prior, full pool, 7 days and 1 day	3
Training budget	2 vs 5 epochs, direct pool	3
Drift detection	Regions 22, 12, 29; abrupt (1997–98 El Niño) and gradual (2013–2022) shifts; four detector settings	4
Expert lifecycle	Regions 22, 12, 29, 1979–2022: reactivation vs spawn, warm start vs fresh, single vs pooled	4
Stage D	Sliding-window PCMCI (1 and 3 seasons) vs ENSO	4

6.4 Metrics

Forecast error is measured by MSE and RMSE $= \sqrt{\text{MSE}}$ in units of the target's standard deviation. Skill is measured against two references:

$$S_{\text{pers}} = 1 - \frac{\text{MSE}}{\text{MSE}_{\text{pers}}}, \qquad R^2_{\text{clim}} = 1 - \frac{\text{MSE}}{\text{MSE}_{\text{clim}}}, \tag{6.1}$$

where persistence predicts $\hat{Y}_{t+L} = Y_t$ and climatology predicts the training mean. At one day persistence is very hard to beat; at seven days OLR keeps little memory, so persistence becomes easy and climatology becomes the demanding reference. Source selection is measured by the number of informative splitters ($\sigma_a \geq 0.02$). Drift detectors are measured by detection latency and false-alarm rate, and the lifecycle by forecast MSE and *forgetting*, the increase in error on a previously seen regime after adaptation. Paired comparisons across the 50 regions use the Wilcoxon signed-rank test; because neighbouring regions are correlated and a single training seed is used, these p-values are indicative.

6.5 Computational Cost

Stage A at 50 regions (900,000 candidates) takes about 23 minutes on one workstation, and the complete Stages A–B, including the 550-node decoy pass and the 596-node second resolution, about 2.5 hours on 20 threads. One 50-region Stage C configuration takes about 20 minutes on 12 worker processes.

Chapter 7

Results and Discussion

This chapter reports the Phase 1 results in the order of the research questions. Every number comes from a saved result file of the project (Appendix A).

7.1 BSISO Season

The BSISO index reproduced from the project's own OLR and u850 anomalies has leading modes explaining 11.2% and 7.7% of the variance, close to the approximately 10% and 7% reported by Lee *et al.* [8]. The slightly larger values are consistent with the smaller domain, whose northern edge is 30°N rather than 40°N. The activity-based season (Figure 7.1) lasts 208 days per year on average, ranges from 82 to 286 days, and begins between April and July. A fixed May–October window would therefore include inactive weeks in some years and miss active weeks in others.

7.2 Stage A: The Discovered Graph (RQ1)

PCMCI at an FDR of 0.05 retains **536 links** into the 50 OLR targets, 10.7 per target, out of 900,000 candidates: the graph is very sparse. Of these, 376 lie within the target's own region, mostly its own OLR at lags of 1–7 days, and 160 come from other regions: OLR 72, u850 37, u200 22, h850 18, PW 11 and SST 0. The outside links act quickly, with a median lag of one day, 90% at two days or less, and none at 25 days or more. Once PCMCI conditions on these fast drivers, slow precursors such as SST carry little *direct* information, because their influence has already passed through the atmospheric fields. Convection in a neighbouring region, together with the low-level flow that carries moisture and the upper-level wave response, precedes convection in the target, which is the expected physical picture of propagating convection.

Figure 7.2 maps the outside drivers. Most connect neighbouring regions; a few longer links carried by wind and height fields span thousands of kilometres within one or two days, at equatorial-wave speeds.

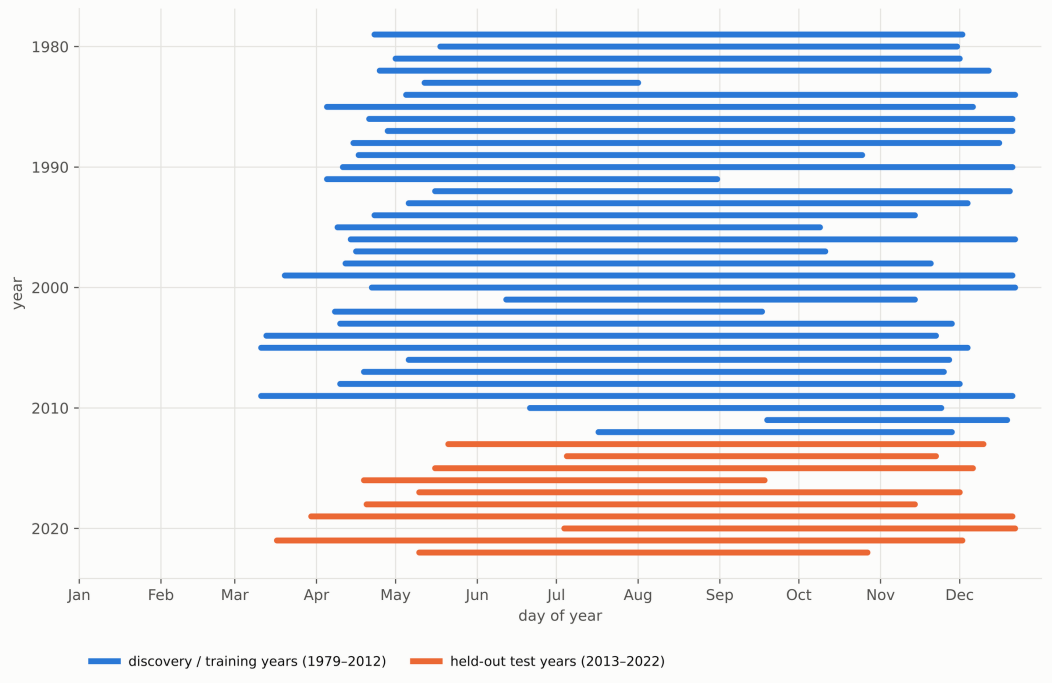


Figure 7.1: Active BSISO season (target days) in each year, derived from the reproduced BSISO index. Blue: discovery and training years (1979–2012); orange: held-out test years (2013–2022).

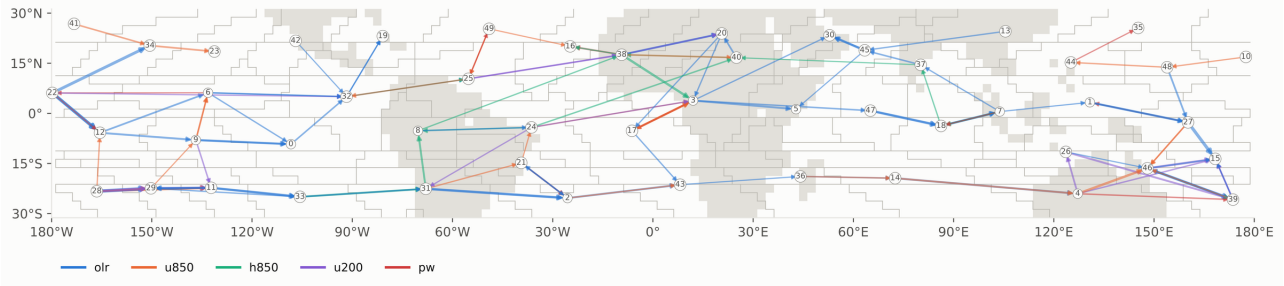


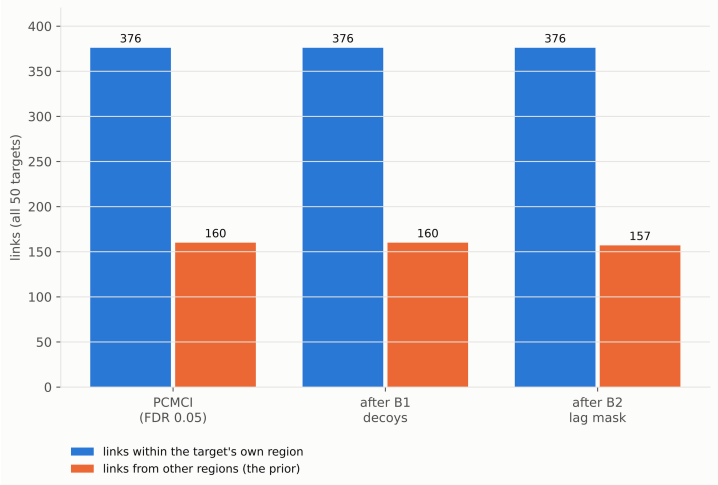
Figure 7.2: Validated outside drivers of the 50 OLR targets (arrows from source region to target region), coloured by the source field.

Worked example: region 22. Region 22 lies over the equatorial western Pacific near the Maritime Continent. PCMCI finds 13 links for it: 10 within the region and three from outside (h850 from region 27 at lags 1 and 3, and PW from region 6 at lag 2). Both source regions are about 280 km away, so the implied speeds of 1–3 m s^{-1} are physically easy. The prior is region 27 (strength 0.21) and region 6 (0.05). Independently, the splitter trained with no prior selects regions 6, 12 and 27, which include both prior regions: two unrelated methods agree on the same drivers. Table 7.1 follows the region's links through the checks.

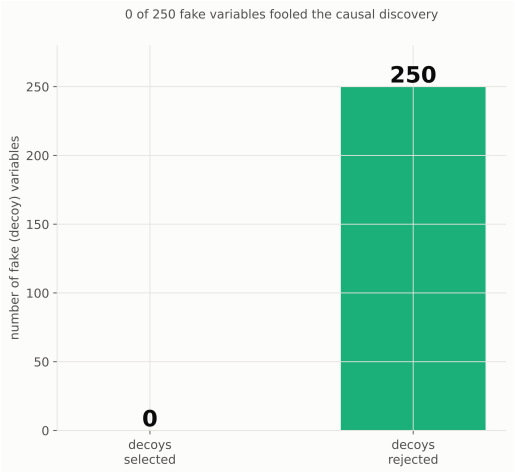
Table 7.1: The links of target region 22 through Stages A and B.

Step	Links	What happened
PCMCI	13	10 own-region (OLR lags 1–3, 5–7; u850, h850 lags 1–2); h850 from region 27 (lags 1, 3); PW from region 6 (lag 2)
B1 decoys	13	no decoy selected anywhere
B2 lag mask	13	sources ≈280 km away; 1–3 m s^{-1}
B3 top-k	3 outside	all three outside links kept
B4 persistence	0.67	both region-27 links persist; region-6 PW link flagged
Prior	2 regions	region 27 (0.21), region 6 (0.05)

7.3 Stage B: Validation (RQ2)



(a) Links after PCMCI, B1 and B2.



(b) B1: decoys selected vs rejected.

Figure 7.3: The graph survives the first two checks almost unchanged. (a) Own-region links (blue) and outside links (orange) after each check. (b) None of the 250 phase-randomised decoys was selected as a parent of any target.

B1: decoys. None of the 250 decoys was selected (decoy rate 0%, against the 30% warning line), while the real graph barely changed (517 real links in the decoy pass versus 536). The pipeline does not select series merely because they share the autocorrelation and the 30–60-day spectral peak of real climate nodes (Figures 7.3 and 7.4).

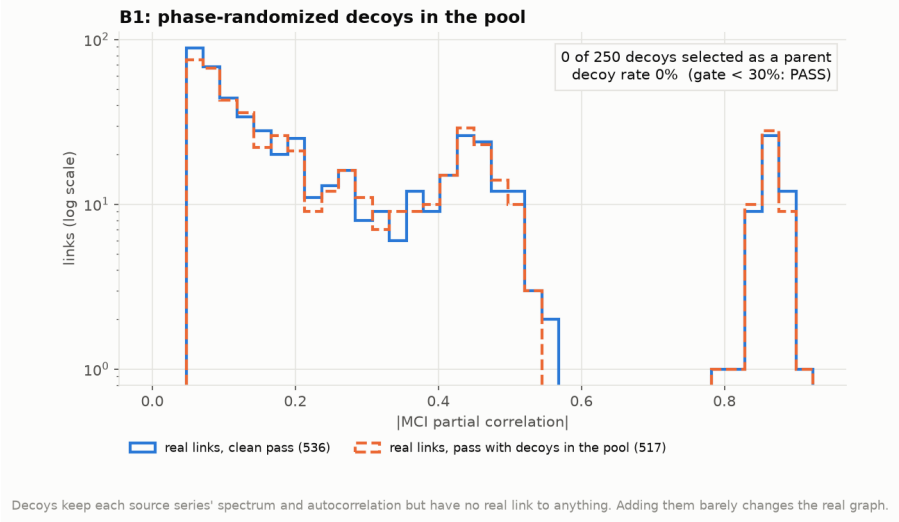


Figure 7.4: B1: distribution of $|\hat{\rho}^{\mathrm{MCI}}|$ for real links in the clean pass (536) and in the pass with 250 decoys in the pool (517). The two distributions nearly coincide, including the strong own-OLR links near 0.85.

B2: physical lag mask. Only **3 of 536** links need an impossible speed: h850 from region 22 to region 32 (5,531 km at lag 1, 64 $\mathrm{m\,s^{-1}}$), and OLR from 28 to 33 (33 $\mathrm{m\,s^{-1}}$) and from 3 to 47 (26 $\mathrm{m\,s^{-1}}$). Of the outside links kept, 28 span 500–5,500 km, and the fastest needs 46 $\mathrm{m\,s^{-1}}$; all of these come from wind and height fields, consistent with equatorial waves (Figure 7.5). No link reaches the 25-day oscillation flag.

B3: top-k. The validation R^2 rises from 0.9619 with own-region links only to 0.9633 with ten outside links, each extra link adding a little. The selected $k = 10$ equals the largest number of outside links of any target, so all 157 surviving outside links are kept: PCMCI with FDR control is already sparse. The prior has **2.06 source regions per target**; two targets (regions 13 and 42) have no outside driver (Figure 7.6).

B4: resolution persistence. At 100 regions the mean persistence score is **0.62**, and **33 of 48** scored targets reach 0.5 (Figure 7.7). OLR links persist best (56 of 70), followed by u200 (14 of 22); moisture and low-level height links, which are more local, persist least (PW 4 of 11, h850 6 of 17).
Hypotheses H1 and H2 are supported: the discovered graph is sparse, physically plausible and largely robust to the spatial partition.

7.4 Source Selection and Forecasting (RQ3)

7.4.1 The splitter without a prior

Without a prior, the expert beats persistence at 7 days in all 50 regions in every candidate pool, with a mean skill of about +0.40. The year-round 7-day forecast in the direct pool has a mean MSE of 0.317 and a mean RMSE of 0.559 standard deviations (Figure 7.9).

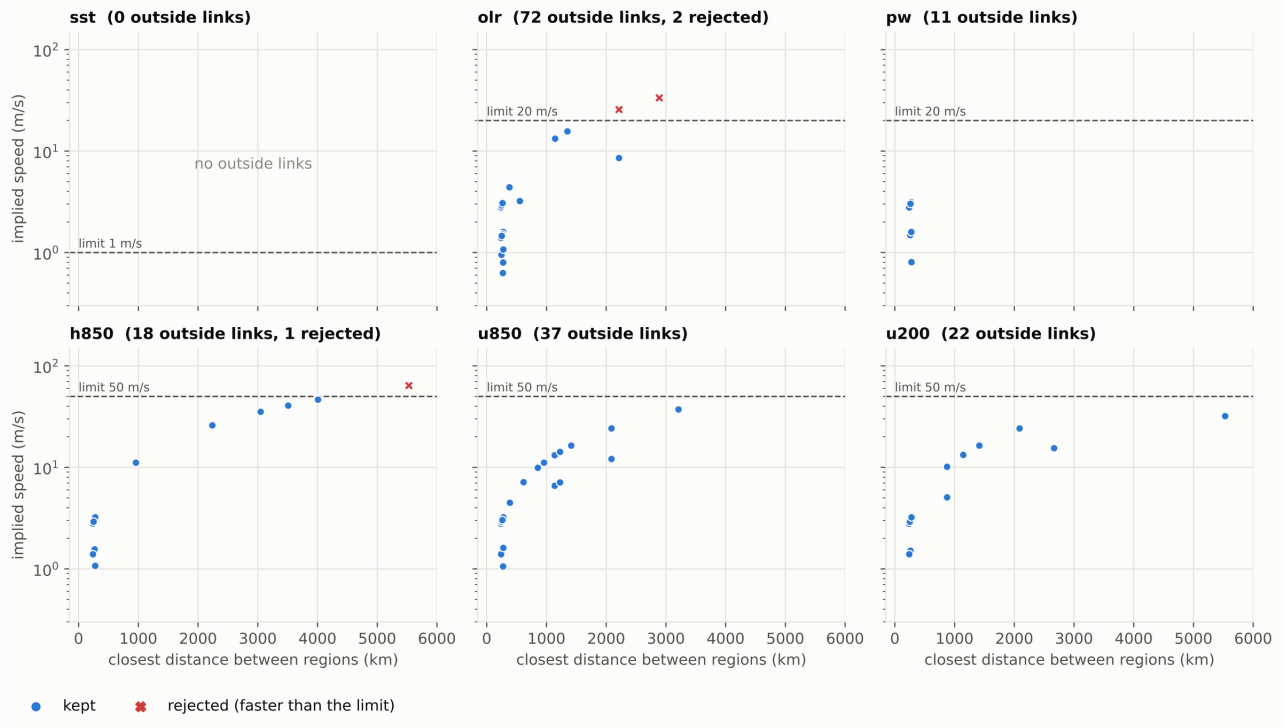


Figure 7.5: B2: implied speed against closest distance for every outside link, by source field, with each field's speed limit (dashed). Crosses are the three rejected links.

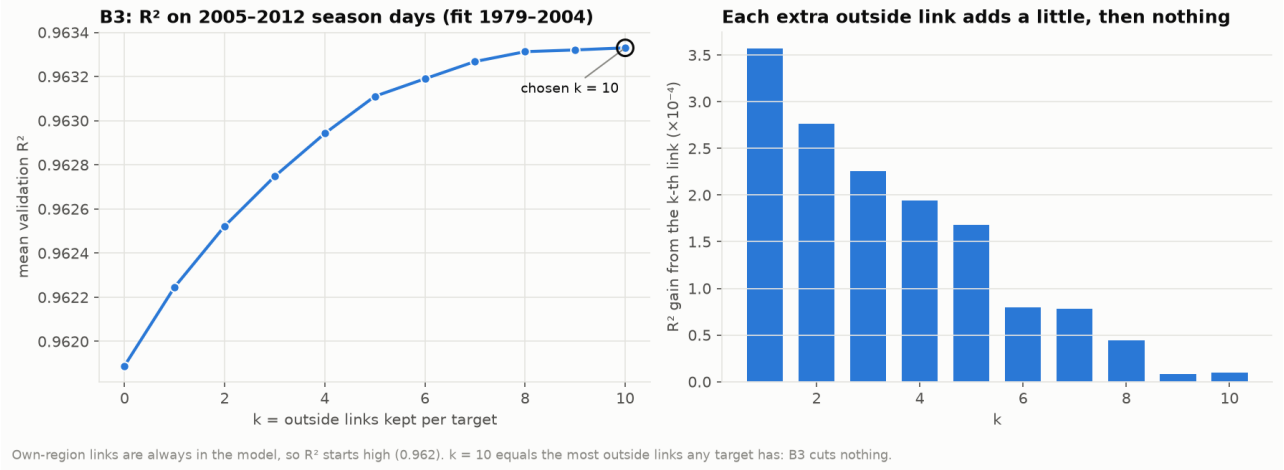


Figure 7.6: B3: mean validation R^2 on 2005–2012 season days against the number k of outside links per target (left), and the gain from the k-th link (right).

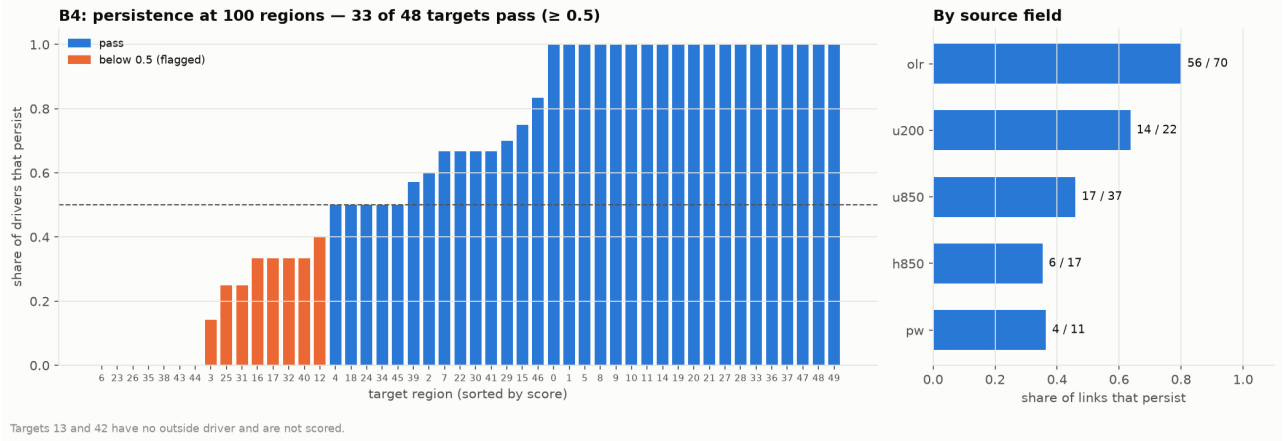


Figure 7.7: B4: persistence score of each target at 100 regions (left; orange below 0.5, flagged) and the share of links that persist by source field (right).

Source discrimination, however, depends strongly on the pool (Table 7.2): 27–30 regions have informative splitters in the direct pool, 8–9 in the 2hop pool, and none in the full pool. With 50 similar-looking candidates, the shared scoring network cannot separate them. Figure 7.8 shows the same contrast: accuracy is flat across pools while discrimination collapses. Discrimination also needs enough training: with the direct pool, 5 epochs give 27 informative regions against 5 after 2 epochs (Figure 7.10a).

Table 7.2: No-prior splitter by candidate pool, 7-day lead, 50 regions, test period 2013–2022.

Target days	Pool	Informative splitters	Beat persistence	Mean S_{pers}
Year-round	direct / 2hop / full	27 / 8 / 0	50 / 50 / 50	+0.401 / +0.395 / +0.396
BSISO season	direct / 2hop / full	30 / 9 / 0	50 / 50 / 50	+0.412 / +0.413 / +0.415

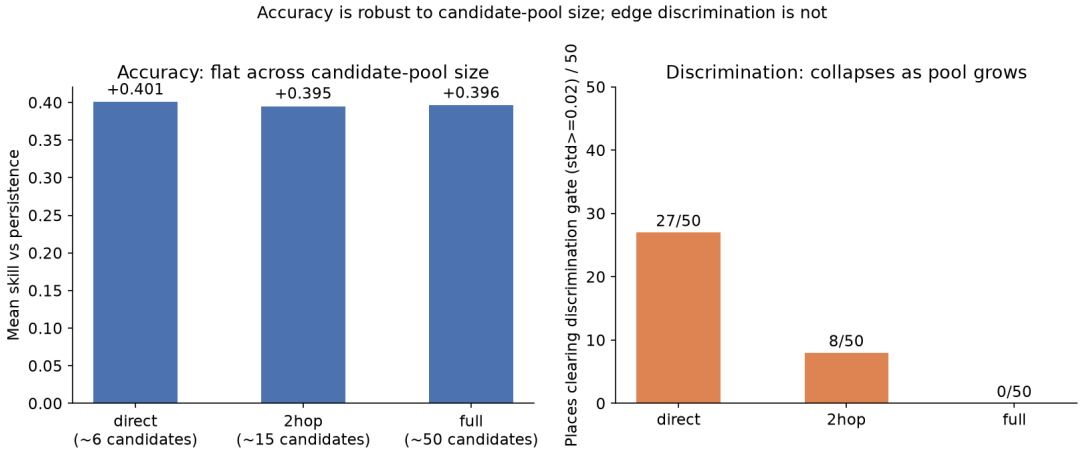


Figure 7.8: No-prior splitter, year-round, 7-day lead: skill relative to persistence is flat across candidate pools (left), while the number of regions with informative splitter scores collapses as the pool grows (right).

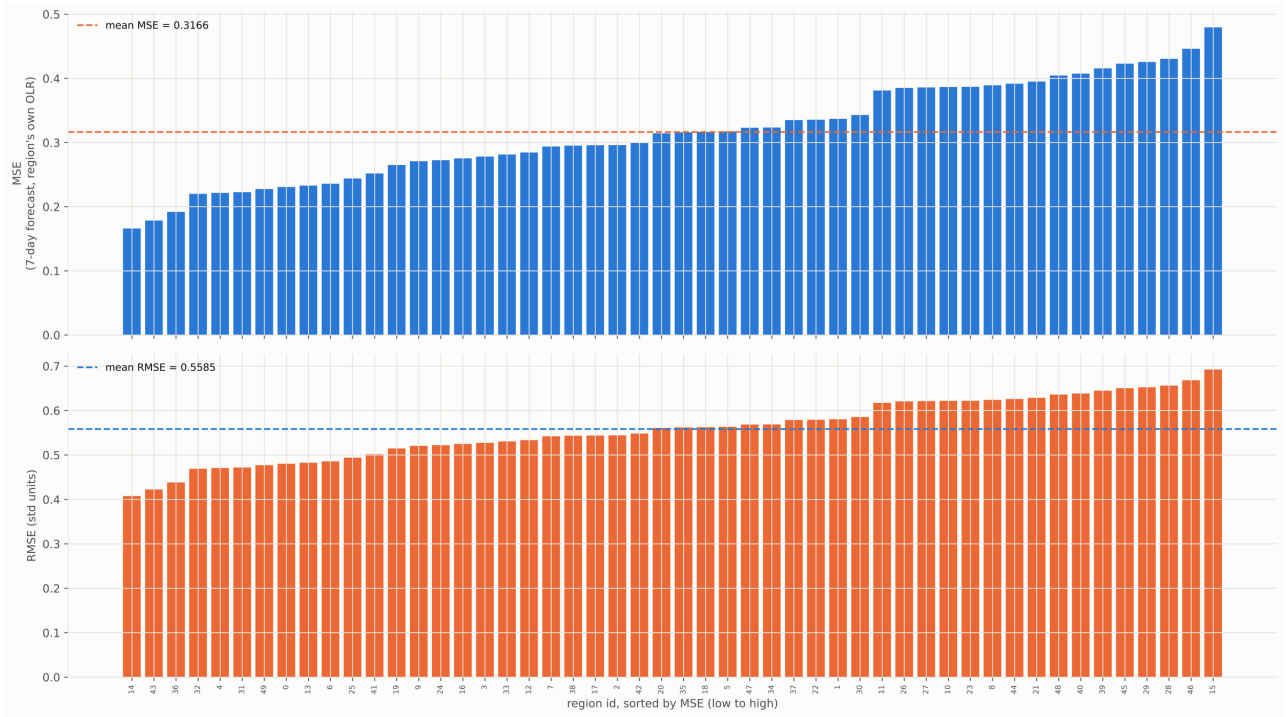
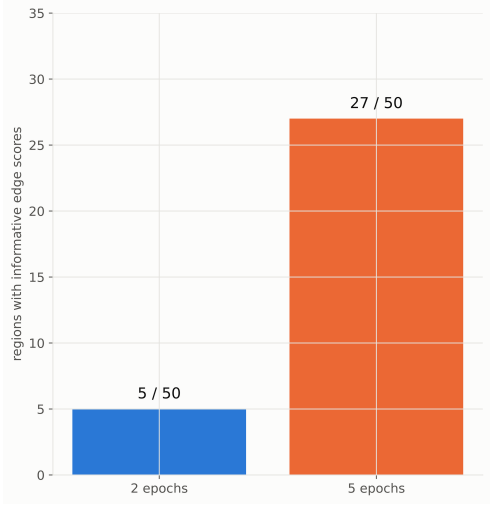
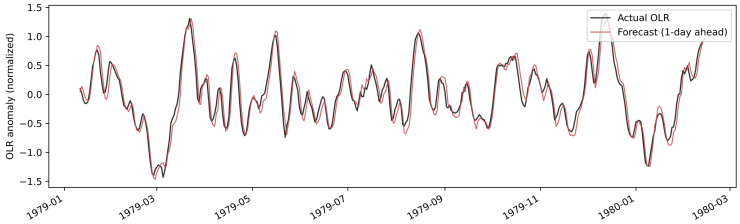


Figure 7.9: Per-region MSE (top) and RMSE (bottom) of the 7-day OLR forecast, direct pool, year-round test 2013–2022, sorted by error. Dashed lines: means (MSE 0.317, RMSE 0.559).



(a) Informative splitters, 2 vs 5 epochs.



(b) Region 22, 1-day forecast vs observed OLR (illustrative 1979–80 window).

Figure 7.10: (a) Splitter discrimination needs an adequate training budget. (b) The expert follows the observed OLR anomalies one day ahead, including sharp swings; held-out scores are reported in Table 7.3.

7.4.2 The causal graph as a prior

Table 7.3 compares the PCMCI prior in restrict mode with no prior in the full pool on the 1,989 held-out season days.

Table 7.3: PCMCI prior (restrict) vs no prior, full 50-region pool, 2013–2022 season days.

Lead	Prior	Informative	Beat pers.	Mean S_{pers}	Beat clim.	Mean R^2_{clim}
7 days	none	0/50	50/50	+0.415	14/50	−0.015
	PCMCI	**24/50**	50/50	+0.416	**19/50**	−0.015
1 day	none	0/50	0/50	−1.527	50/50	+0.720
	PCMCI	**36/50**	3/50	**−1.327**	50/50	**+0.745**

The prior makes source selection informative. In the full pool, where the unguided splitter is uninformative in every region, the restrict prior yields informative splitters in 24 regions at 7 days and 36 at 1 day, without loss of accuracy.

One-day lead: skill improves. The prior improves skill in **32 of 50 regions** (Figure 7.11), with median gains of +0.40 in S_{pers} and +0.048 in R^2_{clim} (Wilcoxon $p = 0.049$) and a median 17% reduction in MSE. Region 22 improves from $S_{\text{pers}} = -0.60$ to -0.05. Skill against persistence is negative at one day for both models because tomorrow's OLR is very close to today's; both are clearly skilful against climatology ($R^2_{\text{clim}} > 0.7$).

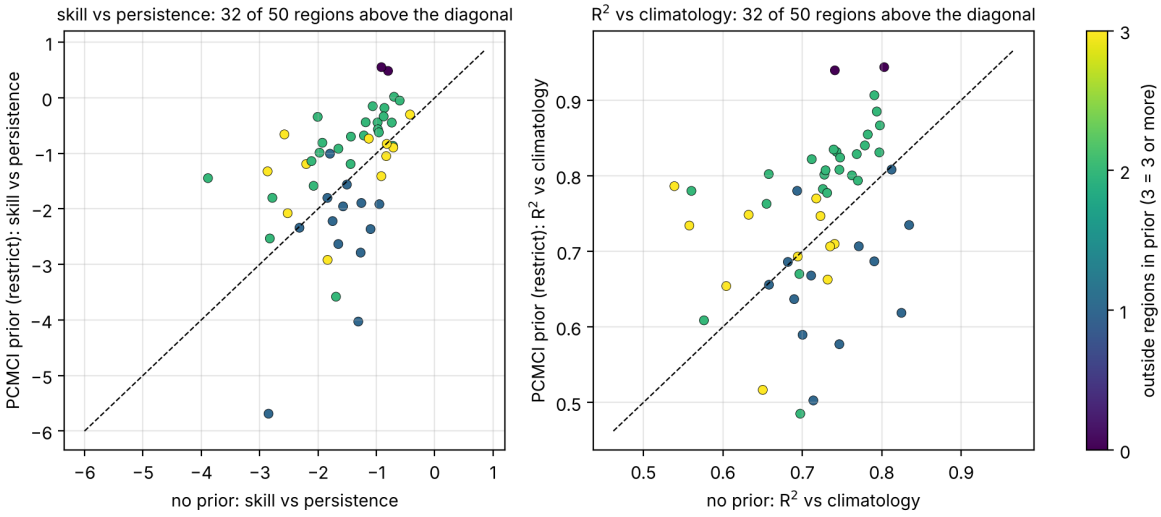


Figure 7.11: One-day lead, full pool: per-region skill with the PCMCI prior against skill without a prior. Points above the diagonal improve; colour gives the number of outside regions in the prior.

Seven-day lead: unchanged. At 7 days, skill is unchanged (23 regions better, 27 worse; mean S_{pers} +0.416 versus +0.415; Figure 7.12), although 19 rather than 14 regions beat climatology. This follows from the graph's time scale: its outside links act within one or two days, so they describe exactly the information a 1-day forecast can use, but a source's state

seven days before the target adds little. The graph identifies the right *neighbours*, while a 7-day forecast needs week-ahead *precursors*.

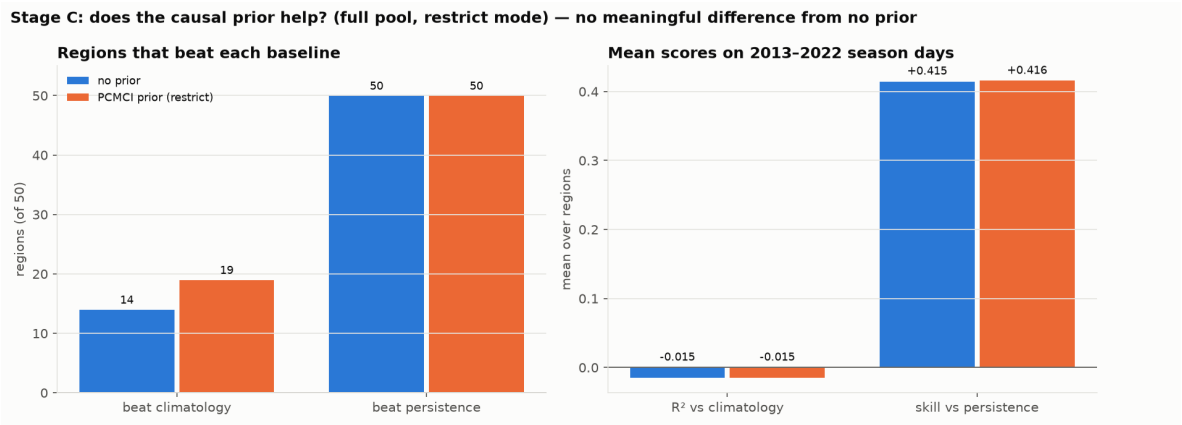


Figure 7.12: Seven-day lead, full pool: regions beating each baseline (left) and mean scores (right), with and without the PCMCI restrict prior.

Dependence on prior size. The one-day gain varies with prior size. The two targets with an empty prior, which are restricted to their own region, improve most (S_{pers} from -0.86 to $+0.52$); targets with two or more prior regions improve on average ($-1.53 \rightarrow -0.91$ for two, $-1.53 \rightarrow -1.22$ for three or more); and targets with a single prior region worsen ($-1.63 \rightarrow -2.48$), because the splitter must then choose between the target's own history and its single driver. Comparing with random priors of equal size, the next step of the experimental programme, will separate the contribution of the graph's choice of regions from that of the smaller pool.

7.5 Drift Detection (RQ4)

Table 7.4 reports the four detector settings on an abrupt shift (the 1997–98 El Niño) and a gradual shift (2013–2022) at three regions.

Table 7.4: Drift detection latency (days from the start of the shift window to the first alarm in it) and false-alarm rate (alarms per year outside all known shift events).

Region	Shift	Ch. 1 error	Ch. 2 structure	Fused OR	Fused AND
22	abrupt 1997–98	159 d	**35 d**	35 d	missed
	gradual 2013–22	966 d	**181 d**	181 d	966 d
29	abrupt 1997–98	223 d	**92 d**	92 d	235 d
	gradual 2013–22	1,158 d	**117 d**	117 d	1,195 d
12	abrupt 1997–98	**95 d**	123 d	95 d	123 d
	gradual 2013–22	**70 d**	100 d	70 d	100 d
False alarms per year (22 / 29 / 12)		0.17 / 0.34 / 1.04	1.62 / 2.25 / 2.19	1.78 / 2.59 / 3.23	0.17 / 0.34 / 0.94

All shifts are detected by the OR fusion at every region. At regions 22 and 29 the causal-structure channel detects the abrupt El Niño shift 2.4–4.5 times sooner than the error channel, and the gradual shift 5.3–9.9 times sooner, which supports H4: watching the causal structure

can reveal a change before it shows in the forecast error. At region 12 the error channel is faster. The structure channel raises more false alarms, so the fusion rule is a genuine trade-off between speed (OR) and caution (AND).

7.6 Expert Lifecycle (RQ4)

Over 1979–2022, the lifecycle handled 49–61 regime changes per region. In **104 of 171** (61%) an archived expert matched the new regime and was reactivated instead of a fresh expert being trained (Figure 7.13): regimes recur, and the archive reuses what was learned about them, supporting H5.

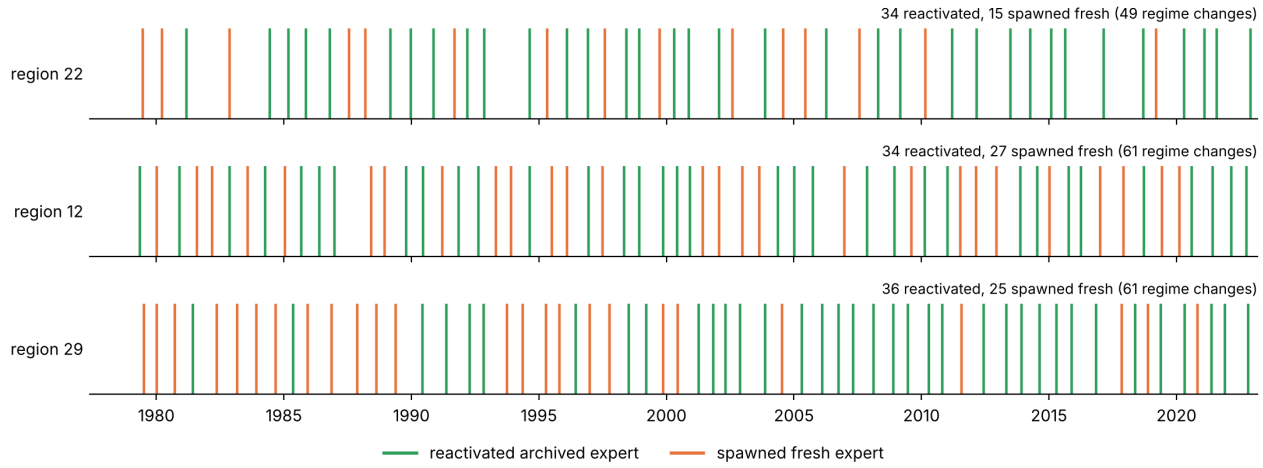


Figure 7.13: Expert lifecycle events, 1979–2022: reactivation of an archived expert (green) or spawning of a fresh expert (orange) at each detected regime change, for regions 22, 12 and 29.

Table 7.5: Reactivation strategies: single best-matching archived expert vs a similarity-weighted blend of the top three.

	Region 22	**Region 12**	**Region 29**
Blend beats single (events)	30/49 (61%)	42/61 (69%)	41/61 (67%)
Mean MSE: single → blend	0.368 → **0.352**	0.422 → **0.402**	0.384 → **0.376**
Forgetting: single → blend	−0.010 → **−0.013**	0.183 → **0.170**	0.092 → **0.086**

Blending the three most similar archived experts beats reactivating the single best match on both forecast error and forgetting at all three regions (Table 7.5). Whether a reactivated expert should be warm-started from its archived weights or trained fresh depends on the region (Figure 7.14): warm-starting gives lower forecast error in 28 of 34 events at region 22 and 24 of 36 at region 29, but lower forgetting only at region 29.

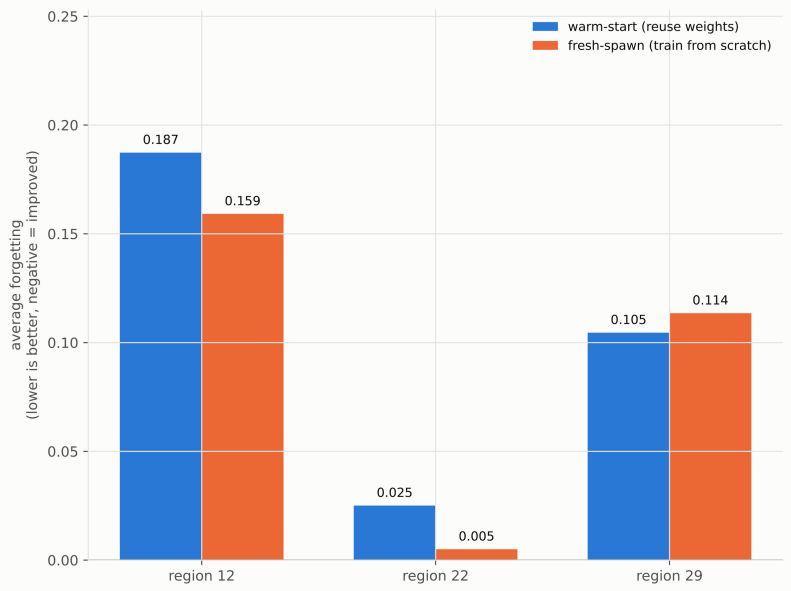


Figure 7.14: Average forgetting (lower is better) when a reactivated expert is warm-started from its archived weights or trained fresh.

7.7 Stage D: Structural Drift of the PCMCI Graph

A Niño-3.4 proxy computed from the project's own SST matches the known ENSO phase in 22 of 24 labelled years. With one-season windows, the parent-set distance is 0.164 when the ENSO phase changes and 0.151 when it does not (Mann–Whitney $p = 0.20$); with three-season windows it is 0.200 versus 0.207 ($p = 0.76$). Each window has only one or two parents per target, so set membership changes mainly through sampling noise (Figure 7.15).

Unlike the splitter-score signal of Section 7.5, the discrete PCMCI parent set is therefore not a reliable regime signal at these window lengths; longer windows and strength-based distances are the natural next step.

7.8 Summary of Findings

Table 7.6: Outcome of the hypotheses.

Hypothesis	Evidence	Outcome
H1 Credibility	0/250 decoys; 3/536 speed violations	Supported
H2 Resolution persistence	33/48 targets ≥ 0.5; mean 0.62	Supported
H3 Graph prior helps	Informative splitters $0 \rightarrow 24$ (7 d) and $0 \rightarrow 36$ (1 d); 1-day skill better in 32/50 regions; 7-day skill unchanged	Supported at 1 day
H4 Earlier detection	Structure channel 2.4–9.9× faster at regions 22 and 29	Supported at 2 of 3 regions
H5 Regime reuse	104/171 changes reactivated an archived expert; blend best	Supported

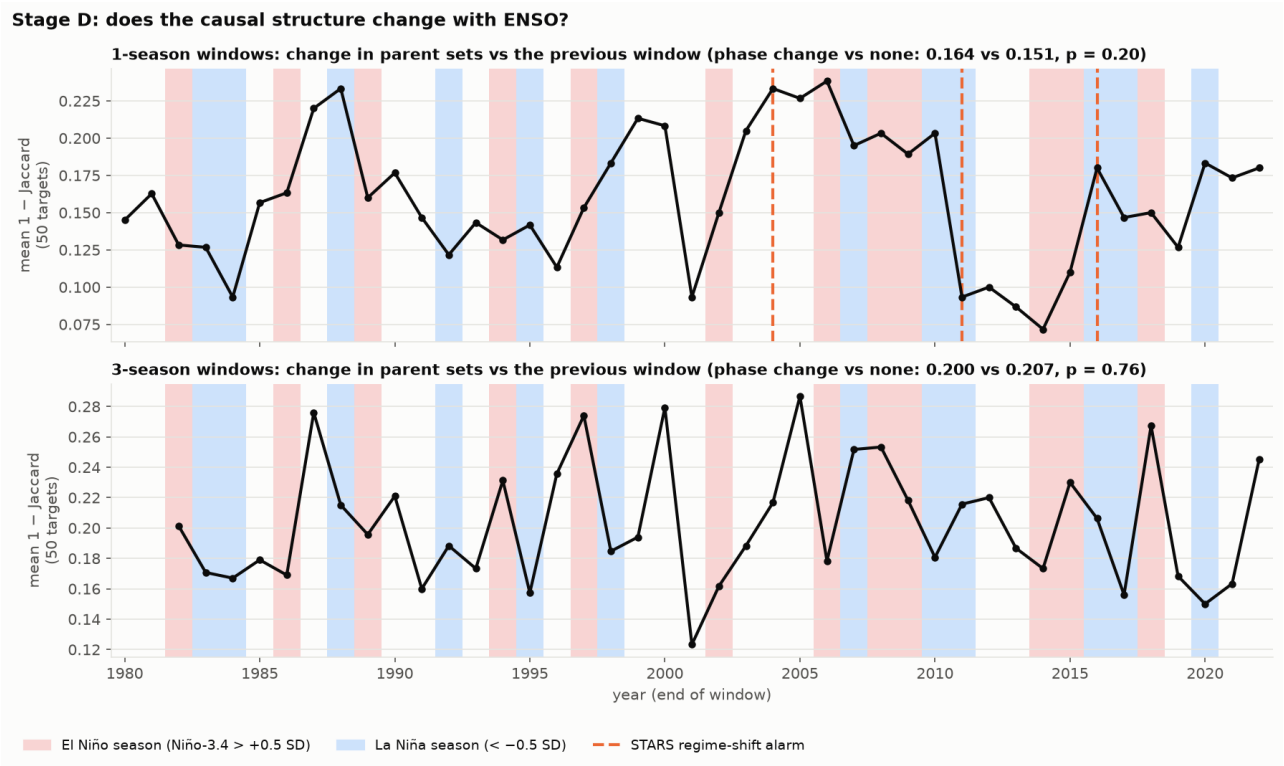


Figure 7.15: Stage D: mean parent-set distance between consecutive one-season (top) and three-season (bottom) windows, with El Niño (red) and La Niña (blue) seasons shaded and STARS alarms dashed.

7.9 Limitations

- **No ground truth.** The checks can reject a bad graph but cannot prove a graph correct; every causal claim is a statement of conditional dependence between region means.
- **Hidden drivers and power.** PCMCI assumes all common drivers are observed; BSISO phase, ENSO and moisture are only partly captured. With 18,000 candidates per target, some links are missed and some false links remain.
- **Time-scale match.** The validated links act within one or two days, shorter than the 7-day lead; a graph built with lags of at least seven days is needed for week-ahead guidance.
- **Evaluation scope.** The forecasting comparisons use a single training seed and compare against the no-prior model; the drift and lifecycle results cover three regions. The upstream climatology (1979–2014) overlaps the test period by two years.
- **Absolute skill at 7 days.** At seven days the models beat persistence everywhere but not climatology on average ($R^2_{\text{clim}} \approx -0.015$).

7.10 Implications for Phase 2

Phase 2 keeps the validation pipeline (decoys, speed mask, persistence), the activity-based season and the region-level prior; builds graphs with lags matched to the forecast lead; uses

the dual-channel detector with continuous structure signals; and adopts pooled reactivation of archived experts as the default lifecycle action.

Chapter 8

Phase 2: India-Focused Forecasting and Adaptive Experts

Phase 1 tested the causal graph and the adaptive experts on a deliberately controlled problem: one convective proxy, daily data and 50 large regions. Phase 2 carries the components that passed those tests into an operationally relevant setting: hourly, kilometre-scale forecasting of near-surface weather over India. This chapter specifies the design; its experiments are future work.

8.1 Motivation

The BSISO studied in Phase 1 drives the active and break spells of the Indian summer monsoon [9], so the large-scale drivers identified there are physically relevant to India. High-resolution regional reanalysis [33] and dense station records [34] make hourly verification possible, and near-surface heat, humidity and wind over India have strong societal impact. Their predictability changes with monsoon regime and ENSO state, which makes adaptation to shift a requirement. The Phase 2 horizon of up to 72 hours also matches the one- to two-day time scale of the dependencies found in Phase 1.

8.2 Workflow

Figure 8.1 shows the planned workflow.

1–2. Data and preprocessing. IMDAA provides hourly regional reanalysis at about 12 km as the primary input and target; ERA5 [35] provides the boundary context and candidate large-scale drivers (BSISO and ENSO indices, remote circulation); and the NOAA Integrated Surface Database provides station observations. Fields are regridded, hourly time stamps aligned, and normalisation statistics estimated on training years only. Targets are 2 m temperature and relative humidity, 10 m wind components and mean sea-level pressure on the grid, plus 2 m temperature and dewpoint at stations.

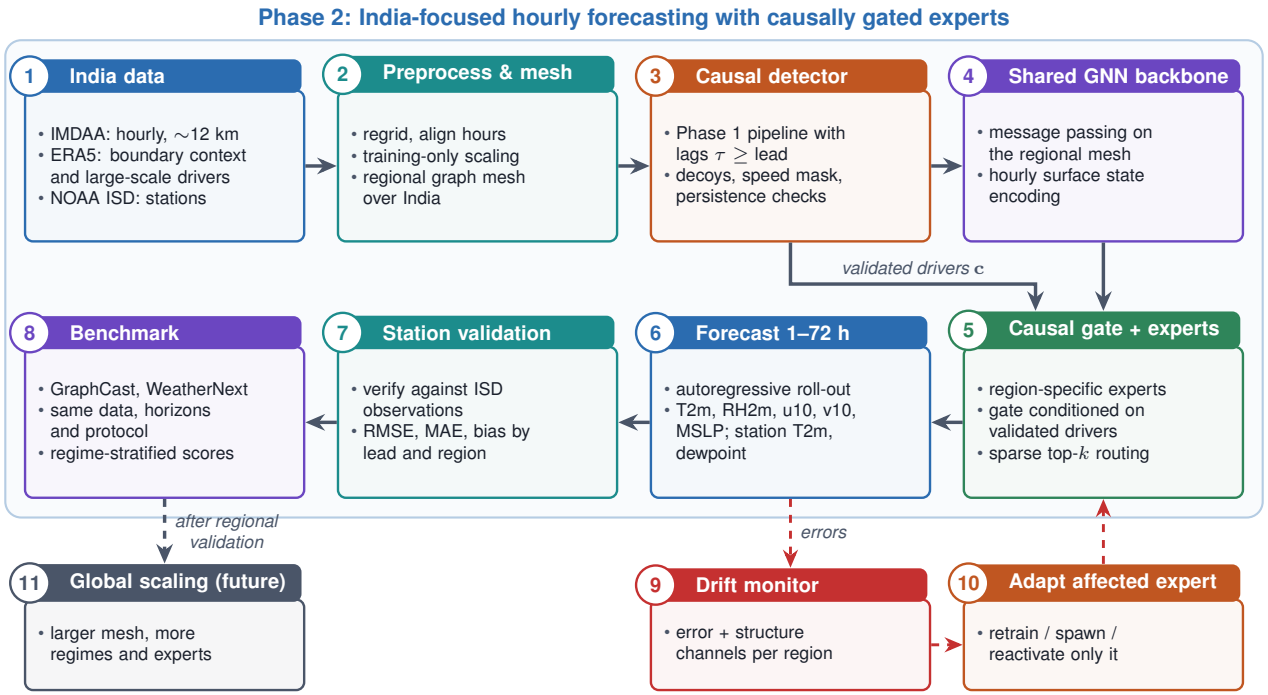


Figure 8.1: Phase 2 workflow. India data are preprocessed onto a regional mesh; a causal detector supplies validated, lead-matched drivers to the gate of a shared GNN backbone with regional experts; forecasts to 72 hours are verified against stations and benchmarked; a per-region drift monitor adapts only the affected expert. Global scaling follows once the regional system is validated.

Table 8.1: Planned Phase 2 datasets (versions, periods and licences to be confirmed).

Dataset	Characteristics	Role
IMDAA [33]	Regional reanalysis for the Indian monsoon region; ∼12 km; hourly	Primary gridded inputs and targets
ERA5 [35]	Global reanalysis; ∼31 km; hourly	Boundary context; candidate large-scale drivers
NOAA ISD [34]	Hourly station observations	Station targets and independent verification

3. Causal detector. The Phase 1 pipeline (PCMCI, decoys, speed mask, persistence) is applied with one change suggested by the Phase 1 results: for a forecast at lead L, drivers are tested only at lags $\tau \geq L$, so that every retained link is usable at the issue time.

4–5. Shared backbone, causal gate and regional experts. A shared graph neural network backbone encodes the regional state by message passing on the regional mesh [12, 32]. Region-specific experts refine its output, and a sparse gate [16] weights the experts using the backbone state together with the validated drivers $\mathbf{c}$:

$$\hat{\mathbf{y}}_{n,t+L} = D(\mathbf{z}_{n,t}) + \sum_k g_k(\mathbf{z}_{n,t}, \mathbf{c}_{n,t})\, E_k(\mathbf{z}_{n,t}), \qquad \textstyle\sum_k g_k = 1. \tag{8.1}$$

In Phase 1 the graph selected *predictors*; in Phase 2 it also conditions the *gate*. A driver too slow to add direct predictive information can still identify the regime, for example an active BSISO phase over India, in which a particular expert is reliable.

6–8. Forecasting, validation and benchmarking. Forecasts are rolled out autoregressively to 72 hours, verified against station observations by RMSE, MAE and bias per variable, lead and region, and benchmarked against GraphCast [12] and WeatherNext using the same data, horizons and protocol. Scores are also stratified by monsoon phase and ENSO state.

Table 8.2: Planned Phase 2 baselines and ablations.

Baseline / ablation	**Purpose**
Persistence; same hour yesterday; hourly climatology	Reference forecasts for skill scores
Backbone only	Value of the experts and gate
Gate without causal context; random or correlation context	Value and specificity of the causal drivers
Full system without adaptation; fixed-schedule retraining	Value of drift-triggered adaptation
GraphCast, WeatherNext	External benchmarks on the same data and protocol

9–10. Drift monitoring and selective adaptation. Each region runs the dual-channel detector of Phase 1. Because the PCMCI parent-set distance proved noisy over short windows, the structure channel uses continuous signals: the splitter-score pattern, which detected shifts early in Phase 1, or strength-based distances over longer windows. Only the affected region's expert is retrained, spawned or reactivated, with pooled reactivation as the default.

8.3 Evaluation under Distribution Shift

The evaluation keeps the Phase 1 discipline of a chronological holdout, frozen graphs and matched controls (a random or correlation context in place of the causal one). It adds regime-stratified scores (El Niño, La Niña and neutral years; monsoon onset, active and break phases),

leave-one-regime-out tests, and adaptation metrics: detection latency, false-alarm rate, error after adaptation, forgetting, and compute cost compared with full retraining.

8.4 Status and Path to Global Scale

The causal-discovery and validation pipeline, the drift channels and the expert archive with pooled reactivation are implemented and tested on Phase 1 data. The India datasets, the shared backbone and the causal gate are designed, and their open choices (backbone architecture, and regional, per-node or regime experts) will be settled on validation data. Once the regional system is competitive, the same architecture extends to global scale: larger meshes, more diverse regimes, long-range teleconnections and a larger expert pool. At that scale causal discovery will need hierarchical candidate sets and the expert pool active management.

Chapter 9

Conclusion and Scope for Further Research

9.1 Summary

This thesis addressed a gap left by existing graph Mixture-of-Experts methods, which route by topology or curvature, keep experts frozen, rely on hand-constructed shift types or add experts on a schedule: none combines causal structure discovery with drift-triggered expert adaptation for spatiotemporal weather forecasting. In Phase 1, a causal graph of lagged climate dependencies was discovered with a verified, vectorised PCMCI over 300 nodes and 60 lags, validated by four independent checks, frozen, and used to guide per-region causal splitters and experts forecasting BSISO-season OLR. A dual-channel drift detector and an expert lifecycle kept each region's expert current without retraining the whole model.

9.2 Achievement of the Objectives

Table 9.1: Objectives and outcomes.

Objective	**Outcome**
O1 Causal graph	536 links from 900,000 candidates at FDR 0.05; short-lag and physically interpretable
O2 Stress tests	0/250 decoys; 3/536 speed violations; $k = 10$; persistence 0.62 (33/48 targets)
O3 Graph as prior	Informative splitters $0 \to 24$ (7 d) and $0 \to 36$ (1 d); 1-day skill better in 32/50 regions
O4 Drift and lifecycle	Structure channel 2.4–9.9× faster at two of three regions; 104/171 regime changes reuse an archived expert; pooled reactivation best
O5–O6 Phase 2	India-focused hourly design, evaluation protocol, benchmark and global path specified

9.3 Principal Findings

1. **The causal graph is trustworthy by every check that can reject it.** No decoy was selected, almost no link needs an impossible speed, most links survive a change of

resolution, and for region 22 the graph and the independently trained splitter select the same drivers.

2. **The graph makes source selection meaningful.** In a large candidate pool, where the unguided splitter cannot discriminate, the graph prior yields informative selections and improves 1-day skill in most regions.
3. **Time scales must match.** The validated links act within one to two days, so the graph helps at short leads and is neutral at 7 days; week-ahead guidance requires a graph built with lags of at least the forecast lead.
4. **Causal structure can reveal drift early.** At two of three regions, the structure channel detected known shifts several times sooner than the error channel.
5. **Regimes recur, and archived experts pay off.** Most regime changes reactivated an archived expert, and blending the best three archived experts reduced both forecast error and forgetting.

9.4 Limitations

The causal claims rest on indirect validation rather than ground truth and assume that common drivers are observed. The forecasting comparisons use one training seed and are made against the no-prior model; the drift and lifecycle results cover three regions; and the data come from a single archive whose climatology overlaps the test years by two years.

9.5 Future Work

1. Compare the causal prior with random and lagged-correlation priors of equal size, and run the initialise and regularise modes and all pools, with several seeds.
2. Rebuild the graph with lags of at least seven days for week-ahead forecasting.
3. Extend the drift and lifecycle evaluation to all 50 regions and to strength-based structural signals.
4. Test the graph on a second reanalysis or OLR product and on a refitted climatology.
5. Implement Phase 2 over India and benchmark it against GraphCast and WeatherNext; then scale to global forecasting.

9.6 Concluding Remarks

Validating causal drivers before using them, and adapting only the regions whose behaviour has measurably changed, gives a forecasting system that is easier to trust and cheaper to keep current than one that is retrained in full. Phase 1 shows that each element of this approach

works on a real tropical forecasting problem and identifies what must change, chiefly matching the graph's time scale to the forecast lead, as the system moves to hourly regional forecasting over India.

Appendix A

Configuration, Reproducibility and Decision Log

A.1 Repository Structure

```
code/causal_moe/
  data/ raw.py, clustering.py, mesh.py, windows.py, splits.py, candidate_edges.py
  causal_graph/ season.py (BSISO index), nodes.py (300/596 nodes), pcmci.py (Stage A),
                validate.py (B1-B4), prior.py (priors and modes)
  splitter/ dirgnn.py (causal splitter)
  experts/ expert.py (per-region expert), pool.py (pooled reactivation)
  drift/ rodionov.py (STARS), channels.py (ADWIN + fusion), archive.py, forgetting.py
code/scripts/ one script per experiment; code/tests/ automated tests
code/results/ result files and figures; thesis/ this LaTeX project
```

A.2 Result Files

Result	**File (under `code/results/`)**
Season, graph, validation, prior	`phase1/bsiso_season.json`, `phase1/stage_a_50.json`, `phase1/b1_decoy_pass.json`, `phase1/stage_a_100.json`, `phase1/causal_graph_50.json`
Pool baselines	`step7_all_places_{direct_lead7,2hop,full}.json`, `phase1/stage_c/*_none.json`
Stage C (7 d, 1 d)	`phase1/stage_c/full_*.json`, `phase1/stage_c_lead1/full_*.json`
Drift detection	`step5_drift_place{22,12,29}_direct.json`
Expert lifecycle	`step6_lifecycle_place{22,12,29}_direct.json`
Stage D	`phase1/stage_d_drift_w{1,3}.json`

A.3 Reproduction

From `code/`, with the pinned environment (`requirements.txt`) and the archive under `data/raw/rt_preprocessed_archive/`:

```
python -m pytest tests
python scripts/build_mesh_cache.py
python scripts/phase1_build_graph.py # Stages A-B (~2.5 h)
python scripts/phase1_run_stage_c.py --pools full --priors none,pcmci --modes restrict
python scripts/run_step5_drift.py # drift channels
python scripts/run_step6_lifecycle.py # expert lifecycle
python scripts/phase1_stage_d_drift.py --window-seasons 1
python scripts/generate_phase1_figures.py
```

The workflow figures are TikZ sources in `thesis/figures/workflows/`; the lifecycle and 1-day Stage C figures are produced by `thesis/scripts/`.

A.4 Decision Log

Decision	**Choice**	**Open alternative**
Prior granularity	Source region (field choice is unstable)	Field- and lag-level prior
Independence test	Linear partial correlation	Nonlinear tests; LPCMCI
Lag range	1–60 days	Lags $\geq$ forecast lead
B2 / B4 failures	B2 removed from prior; B4 flagged	Exclude B4-flagged links
Season	Activity-based, target days only	Full-year graph
Anomaly climatology	Upstream, 1979–2014	Refit on 1979–2012
Phase 2 experts	To be chosen on validation data	Regional, per-node or regime experts

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List of Publications

(Based on the Research Work)

Papers Published

[None at the time of submission. Replace this line with published papers, if any.]

Papers Communicated

[None at the time of submission. Replace this line with communicated papers, if any.]